Week 1

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Deep Learning: Overview

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Deep-Learning

- Deep Learning is an important part of Machine Learning concerned with various algorithms which are inspired by the structure and function of brain called artificial neural networks.

- Deep Learning is considered over traditional machine learning in many instances. The main reason for this is that the deep learning algorithms tries to learn and process very high level of features from data and importantly unstructured data. So this is one of the greatest feature of deep learning over machine learning thus reducing the task of developing new feature extractors for every problem.

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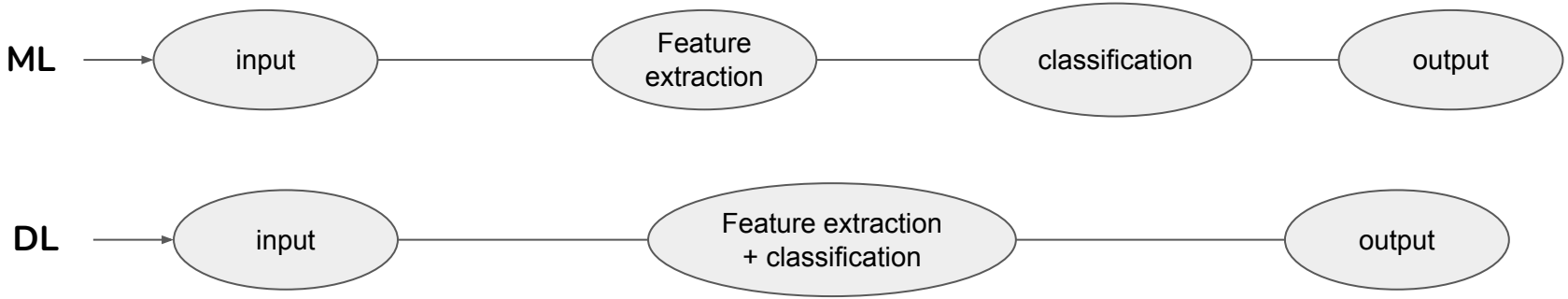
Machine Learning vs Deep Learning

Machine Learning(ML)	Deep Learning(DL)
It can easily train on lesser data and requires lesser time to train	It requires large data and longer time to train
It trains on CPU	It trains on GPU
Some ML algorithms can be easily interpreted	Difficult to interpret.



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Deep Neural Network TOC

1. Introduction to Deep Neural Network
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15. Drawbacks of ANN

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Introduction to DNN

What is Deep Neural Network?

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Deep Neural Network / Artificial Neural Network

1. Deep Neural Network is an Artificial Neural Network with more than one hidden layer.

2. An artificial Neural Network (ANN) is a model that employs a collection of artificial neurons to extract the patterns in the data that represent the relationship between independent and dependent variables.

3. Artificial Neurons are based on the observed behavior of neurons in biological brains. A collection of artificial neurons mimic the behavior of biological neural networks.

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Deep Neural Network / Artificial Neural Network *(continued)*

4. Just as brain uses a network of interconnected neurons to parallelize the processing of input signals to trigger a response, ANN also make use of interconnected neurons to parallelly work on input signals and give an output.

5. In the initial days of the journey towards modern Deep Neural Networks, the research in artificial neuron and ANN were closely tied to the research in neurology. The research in ANN borrowed ideas such as thresholds, linear summation, neuron firing or not etc from Neurology.

6. However, the two fields today are delinked and one does not need to know anything about biological neurons to work on ANN / DNN.

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How it evolved

(History of Artificial Neurons / Neural Networks)

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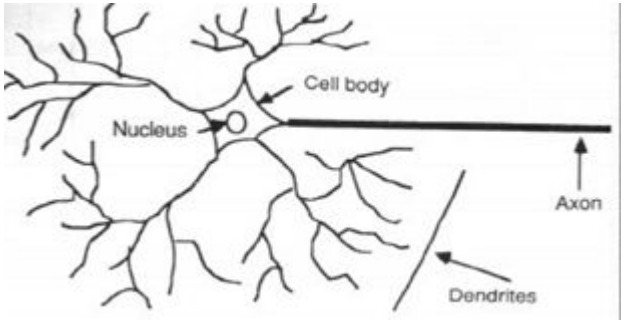
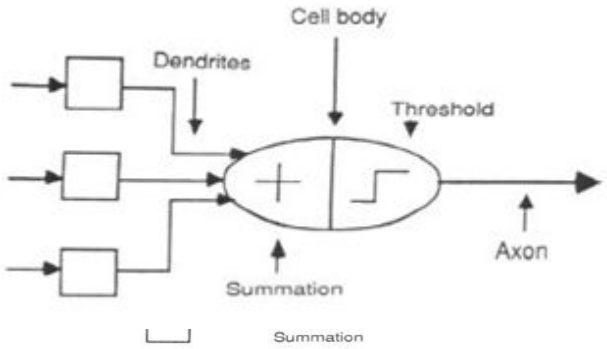
Inspiration from Neurology

Biological Neuron



Artificial Neuron



http://www.andreykurenkov.com/writing/ai/a-brief-history-of-neural-nets-and-deep-learning/

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Boolean Gates and Artificial Intelligence

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Boolean Gates and Artificial Neurons

1. The objective of research in artificial neurons was to develop a computing system that could learn to do tasks on its own without instruction on how to do it.

2. Most of the tasks that we do in our day-to-day life are classified. Hence, research was on to develop an artificial neuron that could classify.

3. Since computing systems are based on Boolean gates which generated two classes, it was natural to check whether the artificial neurons can learn to mimic these gates such as the OR, AND gates.

AND

INPUT		OUTPUT
A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

OR

INPUT		OUTPUT
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

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Warren McCulloch & Walter Pitts Neuron

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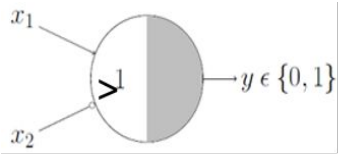
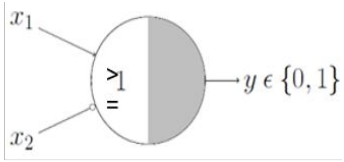
Warren McCulloch & Walter Pitts “AND”, “OR” Neuron



AND



OR

1. In McCulloch Pitt neurons inputs and outputs are binary. The output is only one but inputs can be many

2. All inputs have same positive weights (not shown in the figure)

3. The inputs multiplied with corresponding weights are summed up and the result sent to a step function.

4. The threshold of the step function is fixed (for e.g. 1 for “AND” gate with two inputs each with weight 1)

5. The threshold has to be modified for a gate, for e.g. ‘AND’ gate depending on the number of inputs. What would the threshold for the “AND” gate be if the number of inputs is x1, x2, x3?

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McCulloh Pitts Neuron

AND Gate

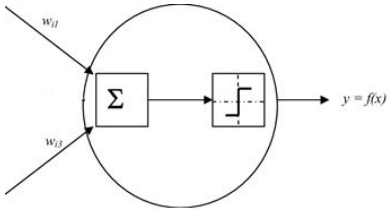
```
w[1] = 1
w[0] = 1
training_data = [
    (array([0,0]), 0),
    (array([0,1]), 0),
    (array([1,0]), 0),
    (array([1,1]), 1), ]

# Step function with threshold of >1. Anything below is 0
step_function = lambda x: 0 if x < 1 else 1

w[1] = 1
w[0] = 1

for x, _ in training_data:
    result = dot(x, w)
    print("{}: {} -> {}".format(x[:2], result, step_function(result)))
```

McCulloh Pitts Neuron



McCullohPitt_Ros
enBlat_Ne

The McCulloch-Pitts model of a neuron is simple. However, this model is so simplistic that it only generates a binary output and also the weight and threshold values are fixed.

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Rosenblatt Neuron Perceptron

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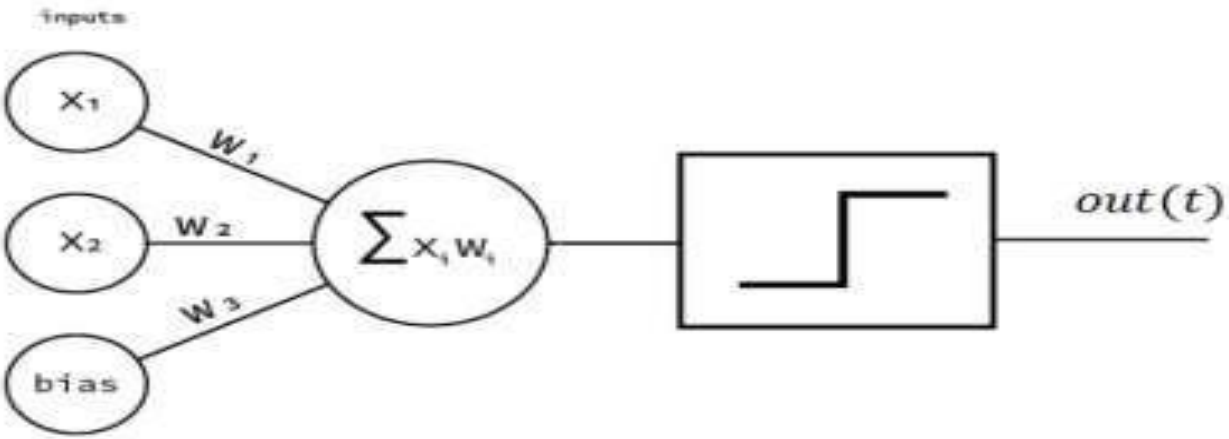
Rosenblatt Neuron / Perceptron



- Uses weights for the inputs. A concept borrowed from William Hebb's rule

  "neurons that wire together, fire together"

- Other than the inputs they also get a special 'bias' input, which just has a value of 1
- Incorrect outputs for a given input are captured as errors

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Rosenblatt Neuron / Perceptron *(continued)*

- A learning rule modifies the weight to correct errors in output
- The neuron could behave like “AND” gate or “OR” gate with same threshold by adjusting weights.
- This neuron learns the patterns from the data. If the data is for “AND” gate, it learns to mimic “AND” gate.
- This concept of learning from data was a major improvement over McCulloch-Pitts neuron.

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Rosenblatt's Artificial Neuron / Perceptron...

- Rosenblatt defined a Perceptron as a simple mathematical model of biological neurons
- It takes inputs as a set of binary values, every input is multiplied by a weight (synaptic strength to each nearby neuron),
- A threshold to evaluate the sum of weighted inputs against. Output a 1 (neuron firing) if the weighted sum is more than the threshold, else a zero (neuron not firing)
- Other than the inputs from sensors / neighboring neurons, they also get a special 'bias' input, which just has a value of 1
- The bias is like an intercept in a linear equation. Useful in generating more functions with the same inputs

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- This was an improvement of the work of Warren McCulloch and Walter Pitts, McCulloch-Pitts neuron.

- McCulloch Pitts neuron had a fixed set of weights associated with inputs and as a result, did not have the ability to learn.

- They had no bias term either Rosenblatt's neuron was inspired by a rule coined by Donald Hebb (researcher studying biological neurons), according to which learning in brain is stored in form of changes to the strength of relationships between two connected neurons "Neurons that fire together, wire together".

- Rosenblatt's neuron can model the basic OR/AND/NOT functions (building blocks of computing systems)

- This was a big step towards the belief that making computers able to perform formal logical reasoning would essentially solve AI.

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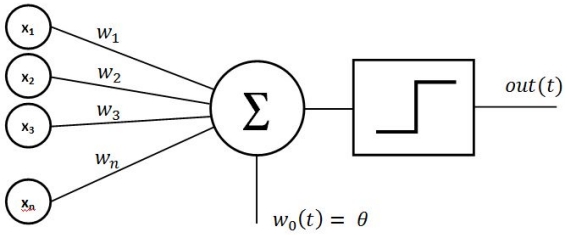
Rosenblatt Neuron / Perceptron



1. Weights for the inputs are not the same, can be positive or negative

2. The output can be -1, 0, 1 unlike MCP neuron where the output is only 0 or 1

3. The neuron is associated with a **learning rule** that modifies the weight to ensure with the same threshold, the neuron can behave like an “AND” gate or “OR” gate with no need for any threshold modification

4. This neuron learns from the data. It has the intelligence to learn the pattern from the data

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Machine Learning (Artificial Neural Network) / Perceptron

Rosenblatt's Neuron Functioning

1. Start with a random set of weights for all the input variables, and the bias.

2. For the input data point, compute the output using the weights and the bias.

3. If the calculated output does not match the expected output (from training data) modify the weights **

4. Go to the next example in the training set and repeat steps j - k until the Perceptron makes no more mistakes.

** The relation between weight adjustment and errors and a learning rate will make the learning possible. Learning is the process finding the right combination of weights that minimizes the errors in the training data set

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Perceptron

Perceptron Learning Algorithm –

1. Select random sample from training set as input. Draw the first random line (green) such that blue triangles lie above it and red circles ones below

2. If the classification is correct, do nothing. But the first time many blue triangles are on the wrong side!

3. If the classification is incorrect, modify the weight vector w and shift the green line

4. Repeat this procedure until the entire training set is classified correctly

5. Howsoever times we run this algorithm, it will find a surface that separates the two classes

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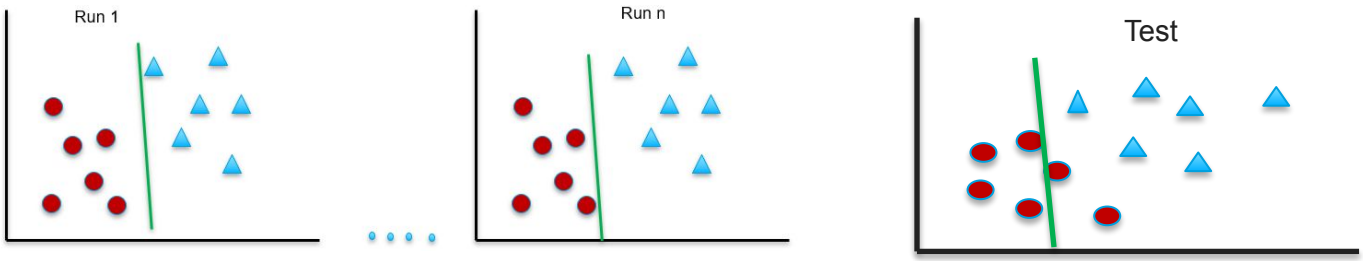

6. The Convergence theorem guarantees that when the classes are linearly separable in the training set, the perceptron will find that surface that separates the two classes correctly

7. The perceptron algorithm does not guarantee it will be able to separate the two classes correctly even when the classes are linearly separable

8. Why? Because it does not look for an optimal plane. It stops the moment it finds the separator plane (a.k.a dichotomizer).

9. Since the planes are passing very close to the data points in the training set, they may not perform well in the test set where the distribution of the data will be different

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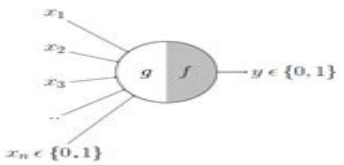
MP Neuron Vs Perceptron

Warren McCulloch & Walter Pitts Neuron

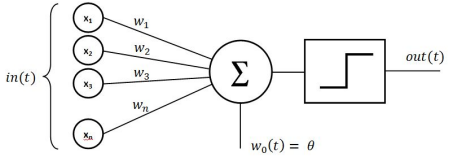


$$g(x_1, x_2, x_3, \ldots, x_n) = g(\mathbf{x}) = \sum_{i=1}^{n} x_i$$

$$y = f(g(\mathbf{x})) = 1 \quad if \quad g(\mathbf{x}) \geq \theta$$
$$= 0 \quad if \quad g(\mathbf{x}) < \theta$$

1. It has an input layer that acts like dendrites.
2. It has two parts, the first part, ***g*** is weighted addition of inputs. The weights are manually initialized and all have same weight.
3. The weighted sum is passed through an activation function ***f*** which yields a 1 if threshold is crossed, else 0.

Rosenblatt's Perceptron



1. It has an Input layer that acts as dendrites
2. It has two parts, first part is weighted addition Each input is multiplied with a weight (which is typically initialized with some random value)
3. The sum is then passed through an activation function which yields a 1 if threshold is crossed
4. The step function can be defined in such a way that output can range from -1 to +1
5. It captured the error and had a logic built in to adjust the weights automatically to reduce the error

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Weakness of Perceptron

1. Papert and Minsky demonstrated that the perceptron was incapable of handling some of the binary gates such as XOR**

2. Given that it cannot represent all possible binary gates, it could not have been used for all possible computations hence the objective of computer based AI was a pipe dream!

3. XOR is an example of distribution of classes not linearly separable in two dimensions but is easily separable in higher 3 dimensional space.Ref: http://www.mind.ilstu.edu/curriculum/artificial_neural_net/xor_problem_and_solution.php

4. It was subsequently demonstrated that instead of making a neuron intelligent, a network of neurons can be used to do what a single neuron could not

5. This was the birth of a Artificial Neural Network

Lab - McCullohPitt_RosenBlat_Neurons.ipynb

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Artificial Neural Network

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Weakness of Perceptron

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Artificial Neural Network

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Origins of Artificial Neural Networks

1. Perceptrons were replaced with artificial neuron which not only had a weighted summation operation but also included a non-linearity function.

2. Multitude of such neurons working together could solve those problems where individual perceptrons failed

3. Multiple neurons in a single layer is akin to transforming data from lower dimensions to higher dimensions! (Cover's theorem in action)

4. Kolgomorov theorem* states that any continuous function f(x1,x2,...,xn) defined on [0,1] with n>2 can be expressed in form of two carefully chosen functions

$$f(x_1, x_2, \ldots, x_n) = \sum_{j=1}^{2n+1} g_j\left(\sum_{i=1}^{n} \phi_{ij}(x_i)\right)$$

5. Looks similar to layer of neurons with non-linear function g acting on summation of weighted inputs ! No wonder NeuralNets form basis of complex processing

* https://en.wikipedia.org/wiki/Universal_approximation_theorem (another mathematical theorem)

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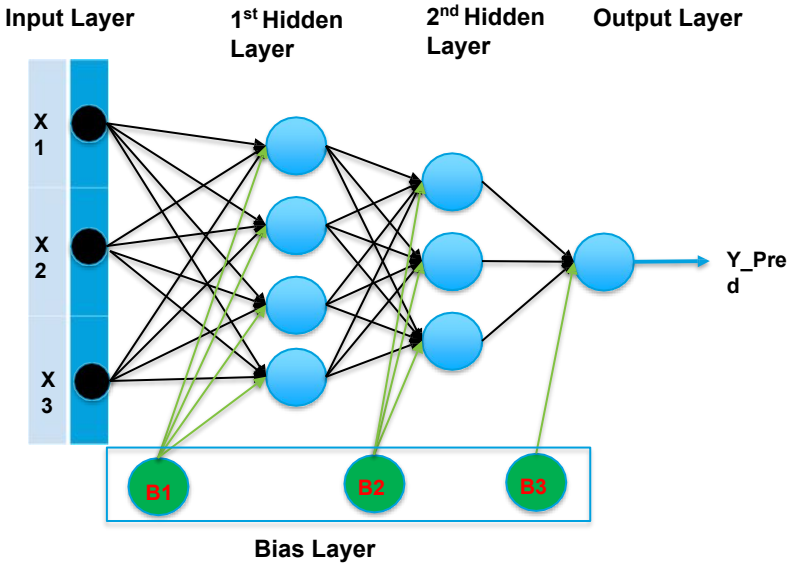
Artificial Neural Network

The processing element of an ANN is called ***a node***, representing the artificial neuron. Each ANN is composed of a collection of nodes grouped in layers. A typical structure is shown The initial layer is the input layer and the last layer is the output layer. In between, we have the hidden layers



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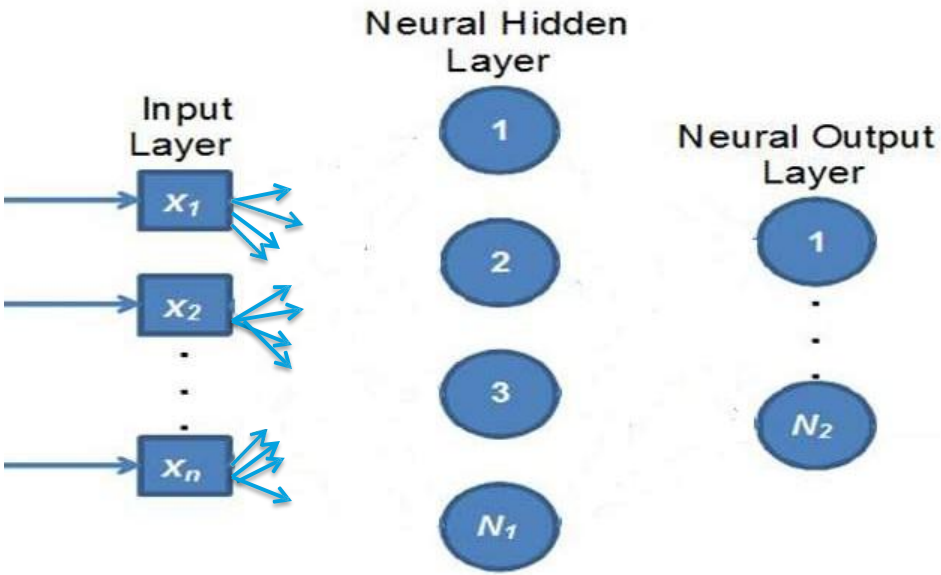
Fully Connected Artificial Neural Network

1.The input layer is passive, does no processing, only holds the input data to supply it to the first hidden layer.



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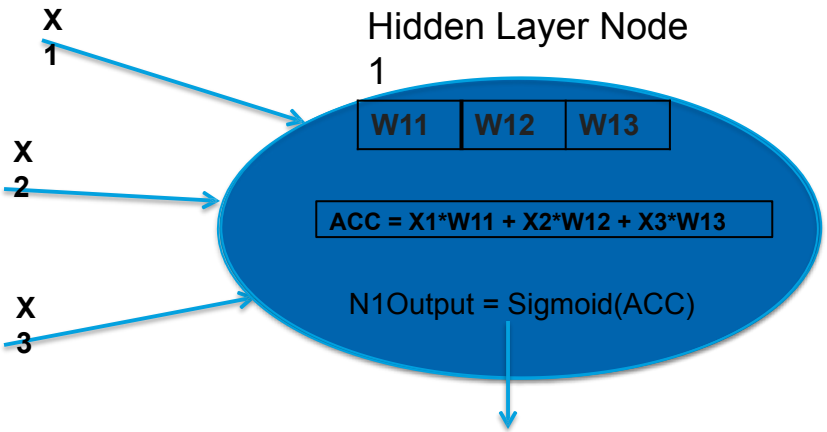
Fully connected Artificial Neural Network

2. Each node in the first hidden layer, takes all input attributes, multiplies with the corresponding weights, adds bias and the output is transformed using non_linear function



3. The weights for a given hidden node is pre-fixed and all the nodes in the hidden layer have their own weights

4. The output of each node is fed to output layer nodes or another set of hidden nodes in another hidden layer

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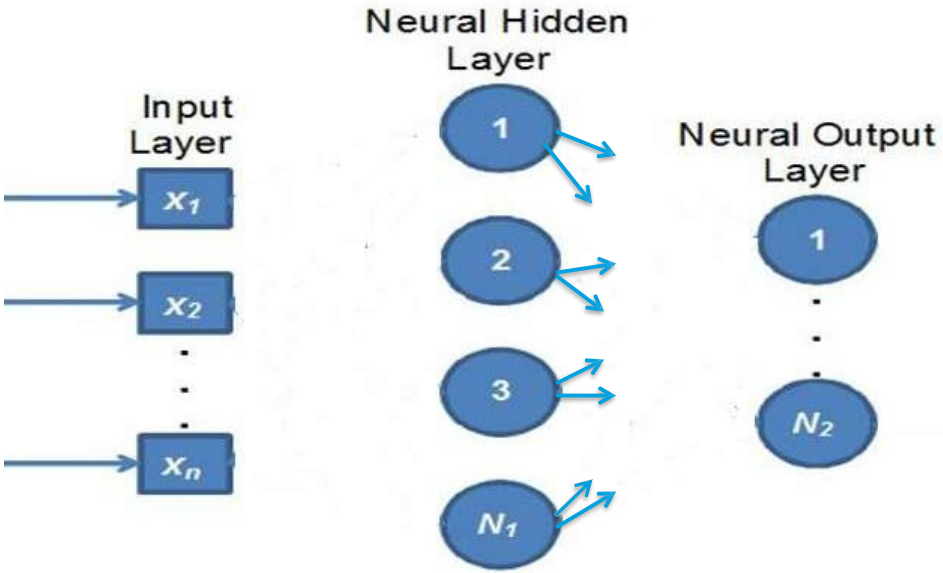
Fully connected Artificial Neural Network

5.The output value of each hidden node is sent to each output node in the output layer



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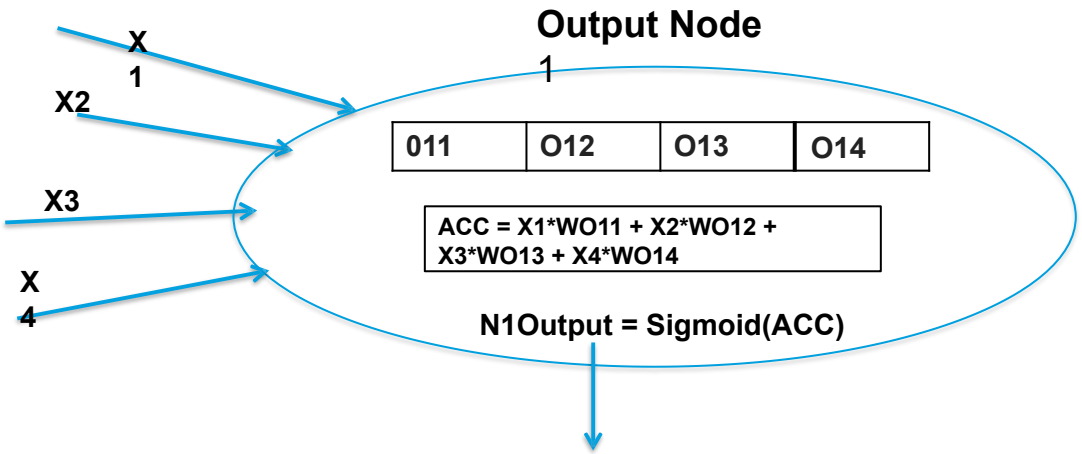
Fully connected Artificial Neural Network



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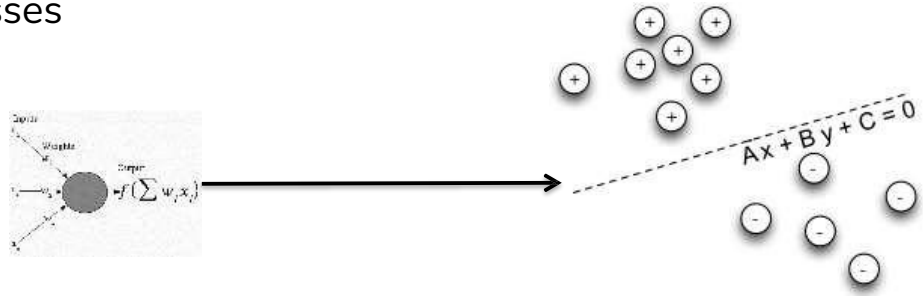
Fully connected Artificial Neural Network

6. In a binary output ANN, the output node acts like a perceptron classifying the input into one of the two classes



7. Examples of such ANN applications would be to detect fraudulent transaction, whether a customer will buy a product given the attributes etc.

8. ANN can also be used for multi-class classification problems such as digit recognition. In fact, it is often used for multi-class classification

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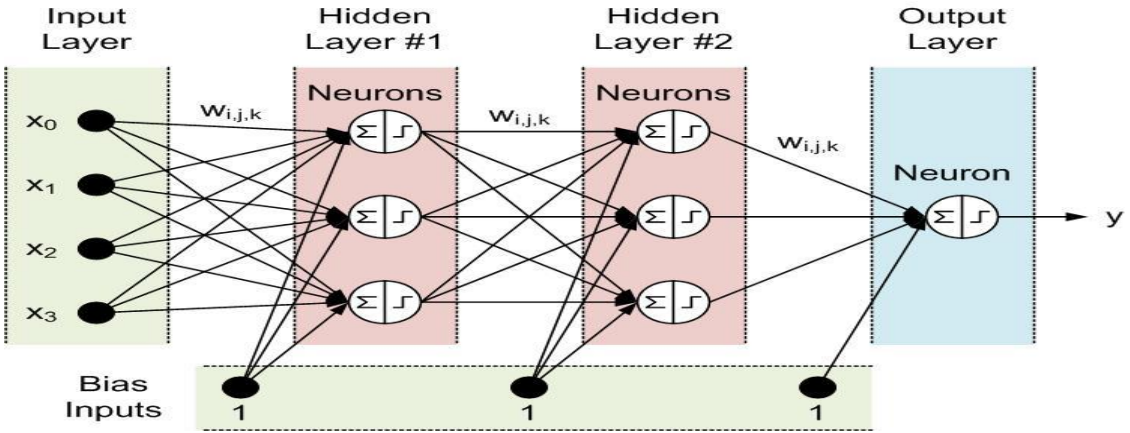
Machine Learning (Artificial Neural Network)

10. The processing elements of a ANN is called ***a node***, representing the artificial neuron. Each ANN is composed of a collection of nodes grouped in layers. A typical structure is shown The initial layer is the input layer and the last layer is the output layer. In between we have the hidden layers



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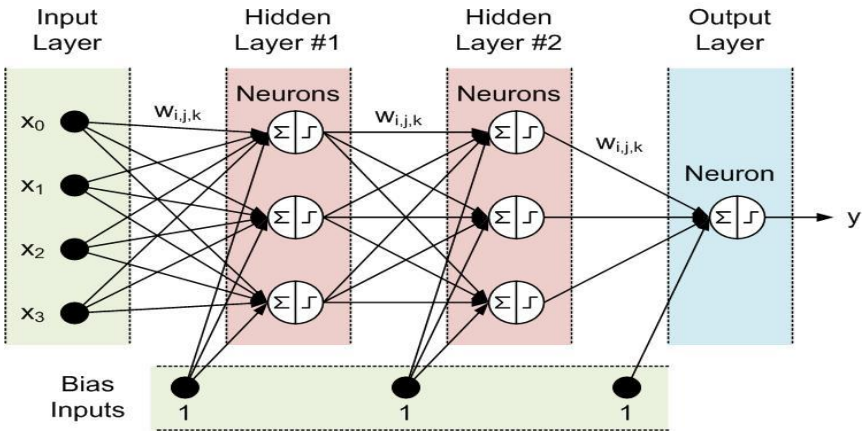
Machine Learning (Artificial Neural Network)

11. Mathematical foundations for artificial neural networks

    a. Kolmogorov theorem – any continuous function f defined on n-dimensional cube is representable by sums and superpositions of continuous functions of only one variable



$$f(x_1, x_2, \ldots, x_n) = \sum_{q=1}^{2n+1} g\left(\sum_{p=1}^{n} \lambda_p \phi_q(x_p)\right)$$

    b. Cover's theorem - states that given a set of training data that is not linearly separable, one can with high probability transform it into a training set that is linearly separable by projecting it into a higher-dimensional space via some non-linear transformation.

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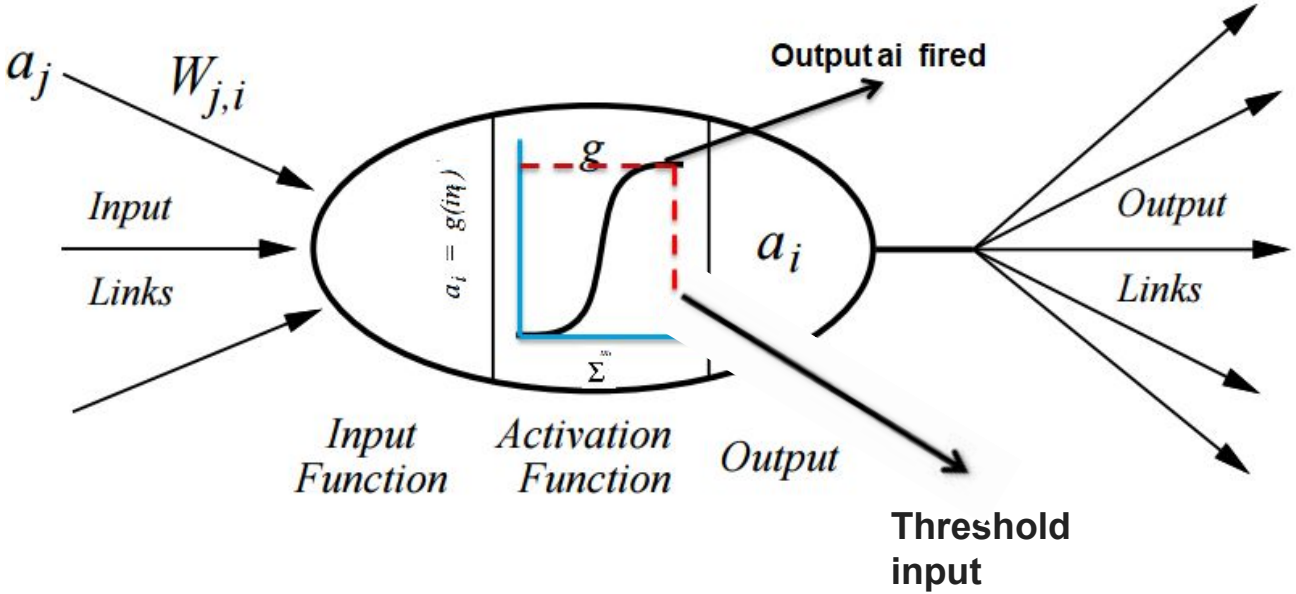
Machine Learning (Artificial Neural Network)

12. A given node will fire and feed a signal to subsequent nodes in next layer only if the non-linear function it implements reaches a threshold.
13. In ANN use of Sigmoid function is more common than step function.



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Application of ANN

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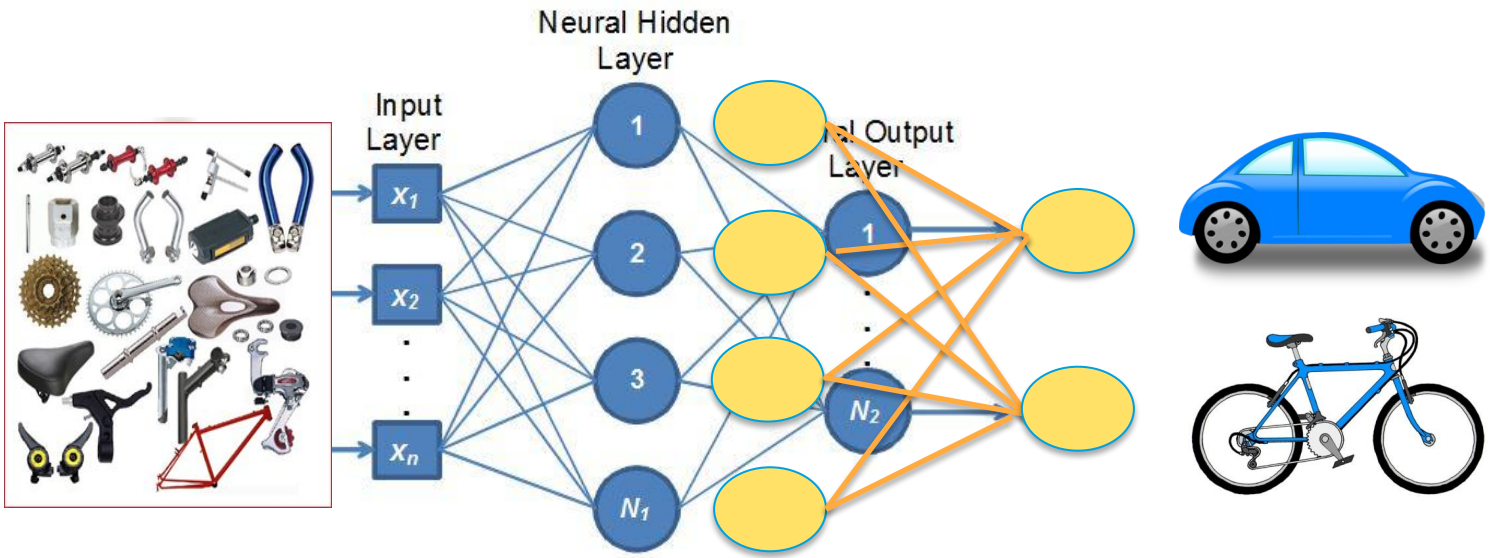
Machine Learning (Artificial Neural Network)

- We can have a ANN with multiple output nodes where a given output node may or may not get triggered given the input and the weights.



- We can have a ANN with multiple output nodes where a given output node may or may not get triggered given the input and the weights.

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Machine Learning (Artificial Neural Network)

- The weights required to make a neural network carry out a particular task are found by a **learning algorithm**, together with **examples** of how the system *should* operate
- The examples in vehicle identification could be a large hadoop file of several millions sample segments such as bicycle, motorcycle, car, bus etc.
- The learning algorithms calculate the appropriate weights for each classification for all nodes at all the levels in the network
- If we consider each input as a dimension then ANN labels different regions in the n-dimensional space. In our example one region is cars, other region is bicycle

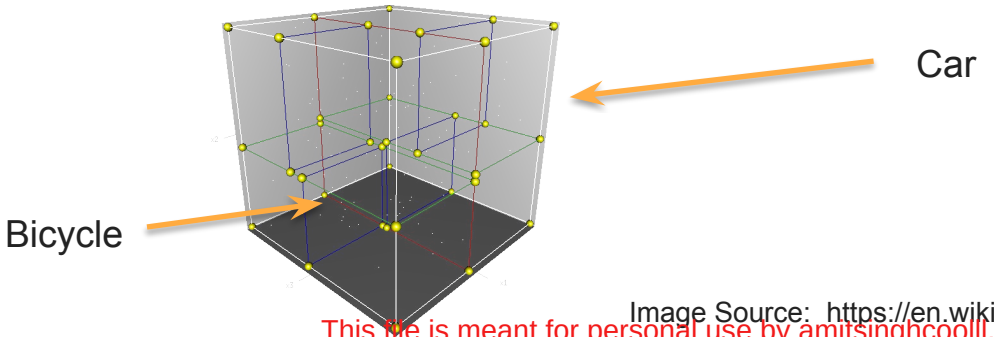

Image Source: https://en.wikipedia.org/wiki/K-d_tree#/media/File:3dtree.png

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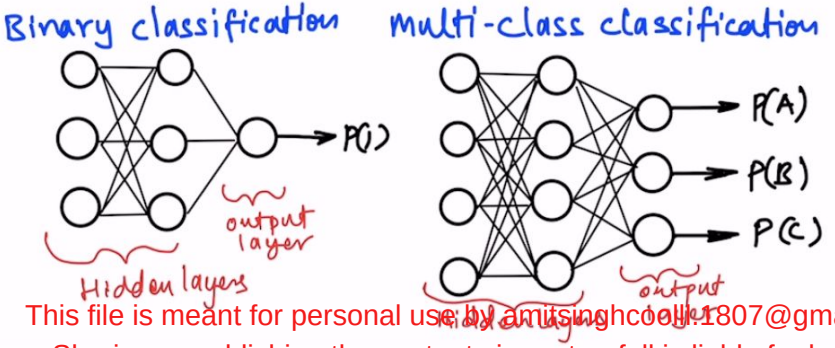
ANN for Classification

Classification always refers to a predictive modeling problem, where the input data is classified into one of the predefined existing classes. In Supervised machine learning the output labels are always known. For example: predicting a patient having cancer or not is an example of binary classification problem.

We have come across many Machine Learning algorithms used for classification but Neural networks can mainly be used when we have many output classes or labels and we have high amount of data for the model.

ANN has many advantages when used for classification but it even has some disadvantages like higher computational costs and longer training time.



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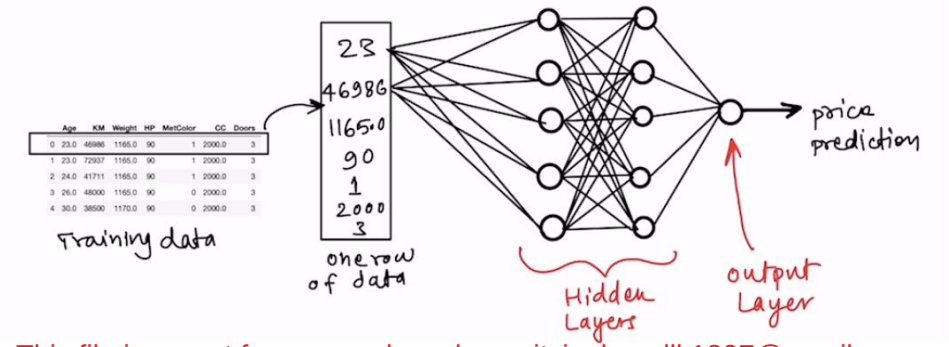
ANN for Regression

- Regression always refers to a predictive modeling problem, where the output values are predicted as a continuous value as a function of inputs.

- Regression models mostly work well when the regression equations fits well on the data. And using ANN is similar to classification but we need to replace the final layer used in the classification model with a fully connected layer with a single node with a single activation function.



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Components of Neural Networks

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Activation Functions

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Activation Functions

1. Artificial neuron works in three steps
    1. First it multiplies the input signals with corresponding weights
    2. Second, adds the weighted signals
    3. Third, converts the result to another value using a mathematical transformation function

2. For the third step, there are multiple mathematical functions available but all together are called the activation function

3. The purpose of the activation function is to act like a switch for the neuron. Should the neuron fire or not.

4. The activation function is critical to the overall functioning of the neural network. Without it, the whole neural network will mathematically become equivalent to one single neuron!

5. Activation function is one of the critical component that gives the neural networks ability to deal with complex problems

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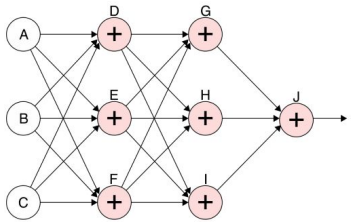
Activation Functions - why?

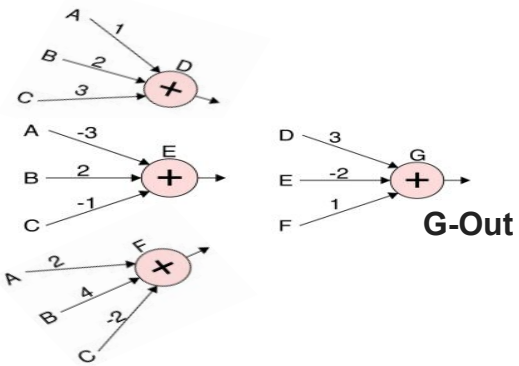
1. Let us take a fully connected neural network. Every neuron in every layer takes multiple inputs
2. The inputs are weighted and summed up at each neuron
3. The nodes in the second layer are simply scaling up the output of neurons in previous layer
4. For e.g the neuron G takes as input weight sums from D,E and F, G's output is scaled version of output of D, E and F
5. G_Out = 3D -2E + 1F

   = 3(1A + 2B + 3C) – 2(-3A + 2B -1C) + 1(2A+4B-2C)

   = <u>8A + 6B + 3C</u>

6. Thus this part of the network is like a single neuron with weights of 8, 6, 3
7. Same argument holds for other neurons
8. Thus the entire neural network collapses to on neuron!
9. A single neuron is not capable of doing much





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Activation Functions - What are they?

1. The activation functions are a mathematical transformations that prevent the network from collapsing to a single neuron

2. The collapse can happen when the neurons do simple addition and multiplications of the inputs. These are called linear operations. Thus linear operations collapse the network

3. All activations functions are nonlinear transformers for exactly the same reason. This non-linear transformation not only prevents collapse, it also empowers the network to do complex tasks because each neuron does something in the network totality

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Activation Functions - Types

4. Types of non-linear activation functions include –
   a. **Piecewise linear functions**
      I. Step function
      II. ReLU – Rectified Linear Units
      III. Leaky ReLU
      IV. Parametric ReLU
      V. Shifted ReLU

   b. **Smooth functions**
      I. Smooth ReLU / Exponential ReLU
      II. Sigmoid / Logistic functions
      III. Hyperbolic Tangent (tanh)
      IV. Swish (combination of Sigmoid and ReLU)

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Activation Functions -

Name	Plot	Equation	Derivative
Identity		$f(x) = x$	$f'(x) = 1$
Binary step		$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for} \quad x \neq 0 \\ ? & \text{for} \quad x = 0 \end{cases}$
Logistic (a.k.a Soft step)		$f(x) = \frac{1}{1+e^{-x}}$	$f'(x) = f(x)(1 - f(x))$
TanH		$f(x) = \tanh(x) = \frac{2}{1+e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$
ArcTan		$f(x) = \tan^{-1}(x)$	$f'(x) = \frac{1}{x^2+1}$
Rectified Linear Unit (ReLU)		$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ x & \text{for} \quad x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$
Parameteric Rectified Linear Unit (PReLU)		$f(x) = \begin{cases} \alpha x & \text{for} \quad x < 0 \\ x & \text{for} \quad x \geq 0 \end{cases}$	$f'(x) = \begin{cases} \alpha & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$
Exponential Linear Unit (ELU)		$f(x) = \begin{cases} \alpha(e^x - 1) & \text{for} \quad x < 0 \\ x & \text{for} \quad x \geq 0 \end{cases}$	$f'(x) = \begin{cases} f(x) + \alpha & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$
SoftPlus		$f(x) = \log_e(1 + e^x)$	$f'(x) = \frac{1}{1+e^{-x}}$

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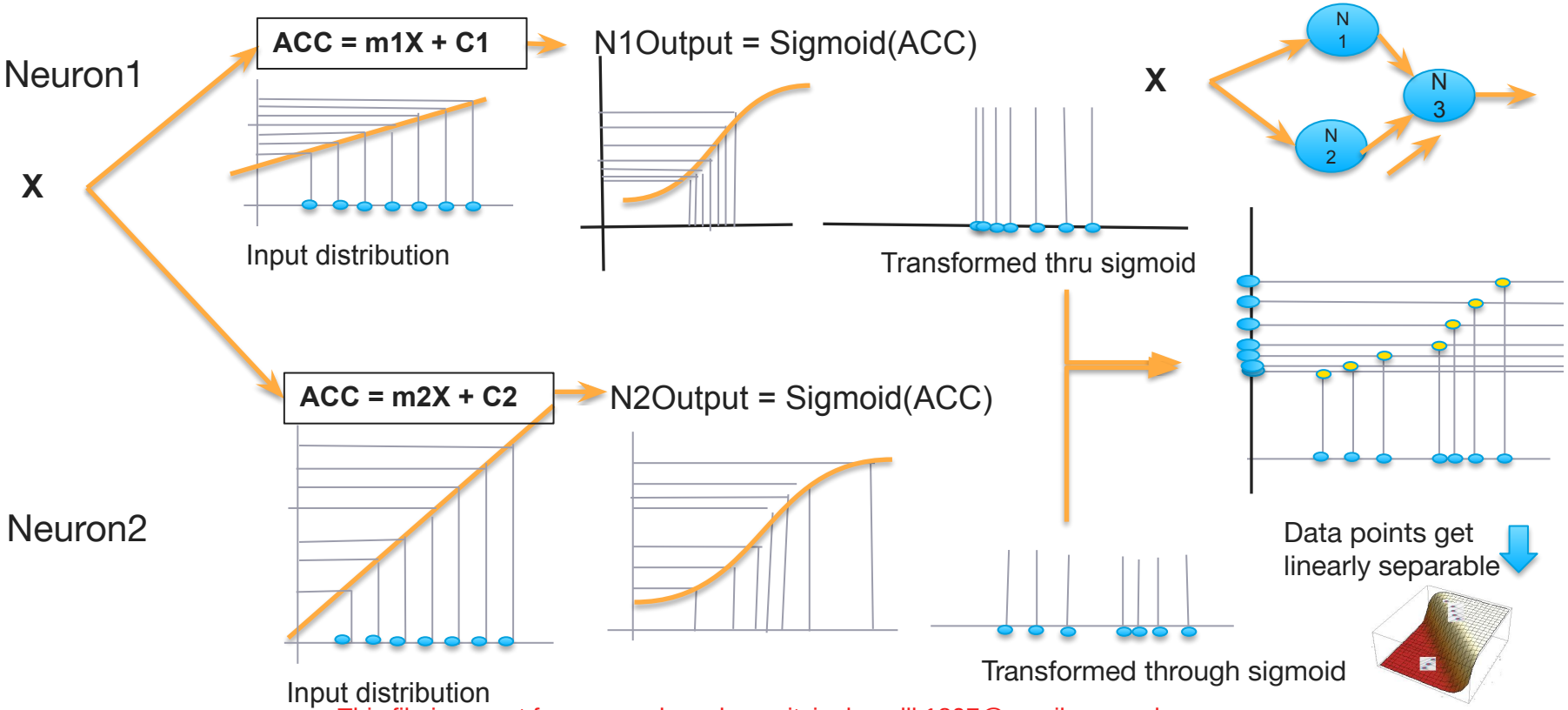
Cover's theorem for non-linear functions



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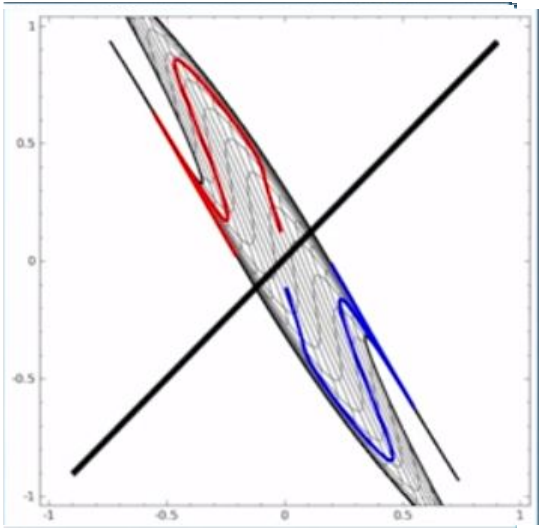
Cover's theorem (Continued)



Neurons stretch the features pace through non-linear functions and achieve Cover's theorem

Image Source: https://colah.github.io/posts/2014-03-NN-Manifolds-Topology/

Ref: https://cs.stanford.edu/people/karpathy/convnetjs//demo/classify2d.html

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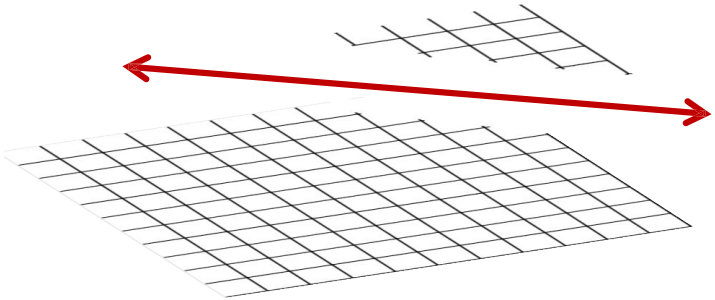
Step Function

Step function divides the input space into two halves 0 and 1



- In a single neuron, step function is a linear binary classifier
- The weights and biases determine where the step will be in n-dimensions
- But, as we shall see later, it gives little information about how to change the weights if we make a mistake
- So, we need a smoother version of a step function
- Enter: the Sigmoid function

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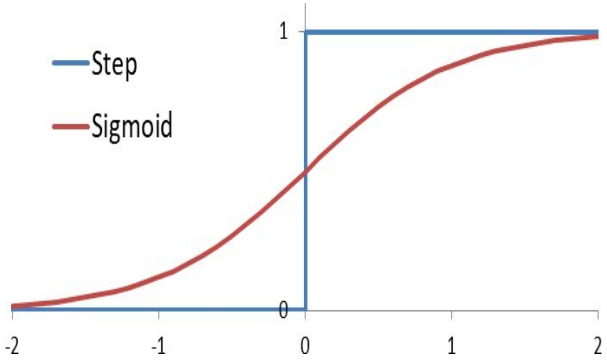
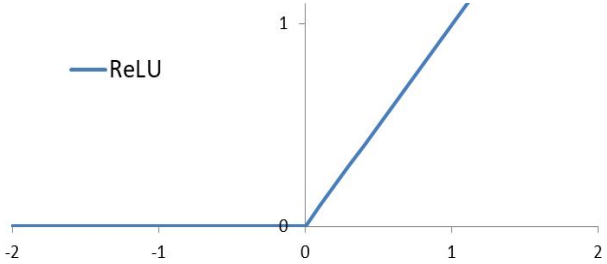
Problems with Activation Functions





- For both large positive and negative input values, sigmoid doesn't change much with change of input.The problem with sigmoid is (near) zero gradient on both extremes
-
- ReLU has a constant gradient for almost half of the inputs
- But, ReLU cannot give a meaningful final output

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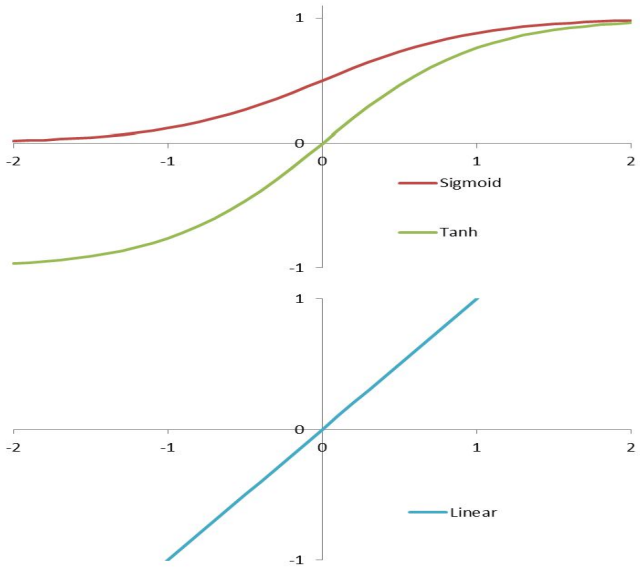
Output activation functions



- Sigmoid gives binary classification output
- Tanh can also do that provided the desired output is in {-1, +1}
- Softmax generalizes sigmoid to n-ary classification
- Linear is used for regression
- ReLU is only used in internal nodes (non- output)

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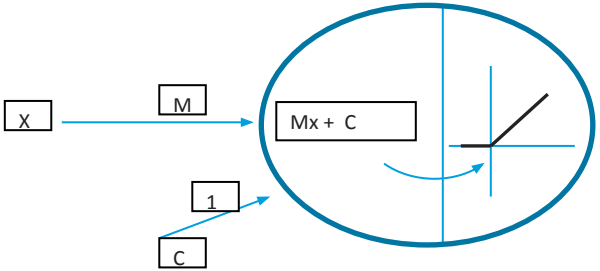
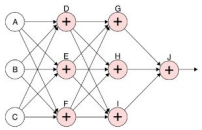
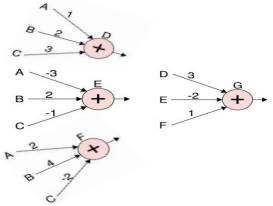
RELU VS Simple linear activation function

1. Half range of RELU (default RELU) is a simple linear transformation of the form y = mx (as long as mx +c >0), for mx + c <=0 y = 0

2. If half of the input is transformed using linear model, then why does not RELU network collapse to Perceptron? To understand this let us look at the dummy network below which has no nonlinear transformation –







3. When we replace all the neurons in the network with the neuron shown above with RELU, Gout will be 0 if 3D -2E + 1F < = 0 and 3D - 2E + 1F will be same as Gout only when 3D -2E + 1F is >0.

4. Thus, Gout cannot be generalized as 3D -2E +1F . Same goes for D0, E0 and F0. Hence the neuron does not become an equivalent of a perceptron

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Softmax Activation Function

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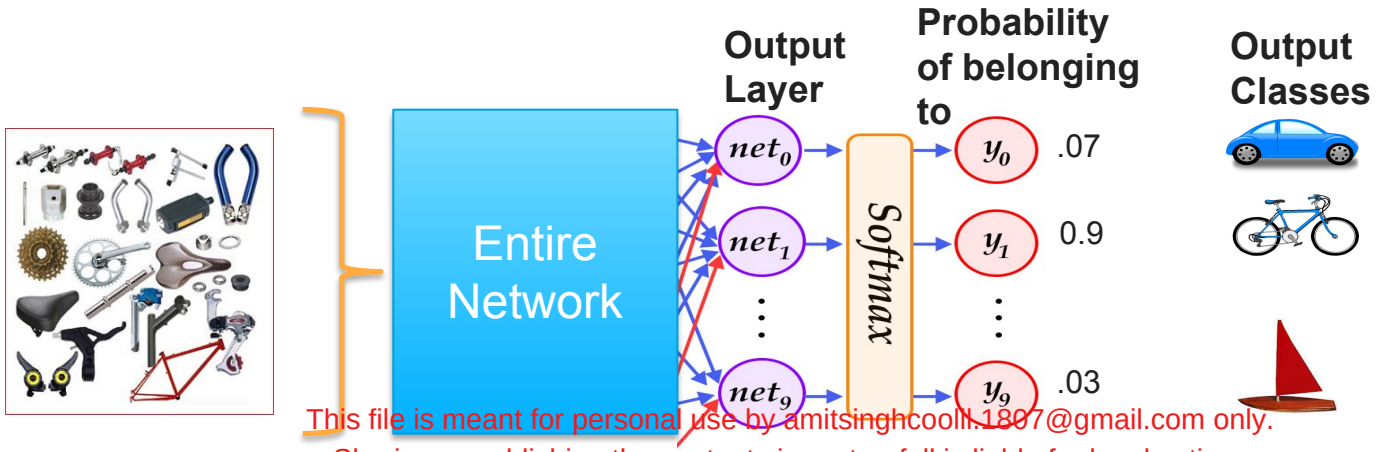
SoftMax Function -

1. A kind of operation applied at the output neurons of a classifier network
2. Used only when we have two or more output neurons and is applied simultaneously to all the output neurons
3. Turns raw numbers coming out of the pen-ultimate layer into probability values in the output layer
4. Suppose output layer neurons emit (Op1, Op2, Opn). The raw numbers may not make much sense. We convert that into probabilities using Softmax which becomes more meaningful. For e.g. input belongs to cycle is 30 times more likely than sailboat, 13 times more probable than car.



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Forward Propagation and Bias

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Forward Propagation

1. The directed acyclic path taken by input data from input layer to get transformed using non-linear functions into final network level outputs

2. Input data is propagated forward from the input layer to hidden layer till it reaches final layer where predictions are emitted

3. At every layer, data gets transformed non-linearly in every neuron

4. There may be multiple hidden layers with multiple neurons in each layer

5. The last layer is the output layer which may have a softmax function (if the network is a multi class classifier)

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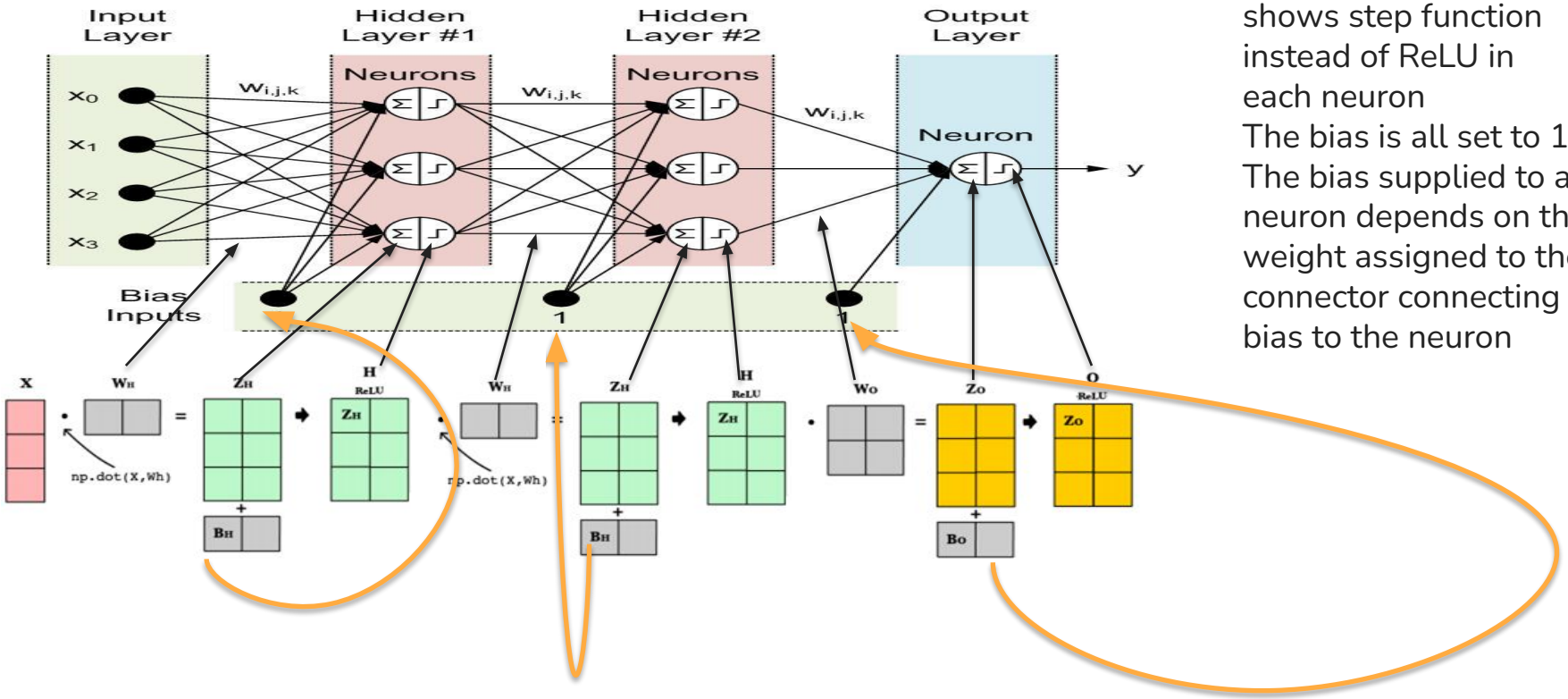
Forward Propagation and Matrix operations



Note: The diagram shows step function instead of ReLU in each neuron
The bias is all set to 1.
The bias supplied to a neuron depends on the weight assigned to the connector connecting bias to the neuron

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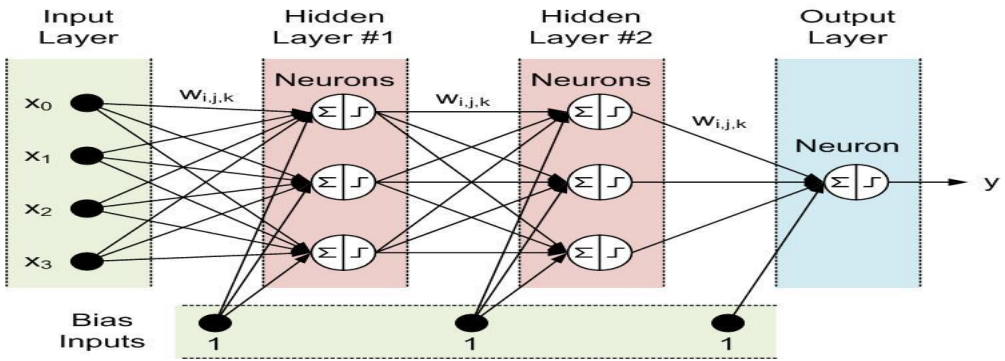
Bias Term

1. Every neuron in the hidden layers are associated with a bias term. The bias term help us to control the firing threshold in each neuron



2. It acts like the intercept in a linear equation (y = sum(mx) + c). If sum(mx) is not crossing the threshold but the neuron needs to fire

3. bias will be adjusted to lower that neuron's threshold to make it fire! Network learns richer set of patterns using bias

4. The bias term is also considered as input though it does not come from data

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Forward Propagation

6. Forward prop steps –
   a. Calculate the weighted input to the hidden layer by multiplying X by the hidden weight Wh
   b. Apply the activation function and pass the result to the final layer
   c. At output layer, repeat step b replacing X by the hidden layer's output

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Loss Function

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Interpreting Bad Model using Loss Function

- Loss trends opposite of accuracy
  - Loss is low when accuracy is high
  - Loss is zero for perfect accuracy (by convention)
  - Loss is high when accuracy is low

- Loss is a function of actual and desired output

- Minimizing the loss function with respect to parameters leads to good parameters

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Properties of a good loss function

- Minimum value for perfect accuracy
  - Usually zero
  - *Note:* low loss on training does not guarantee low loss on validation or testing
- Varies smoothly with input
- Varies smoothly with parameters
- Good to be convex in parameters (but is usually not)
  - Like a paraboloid

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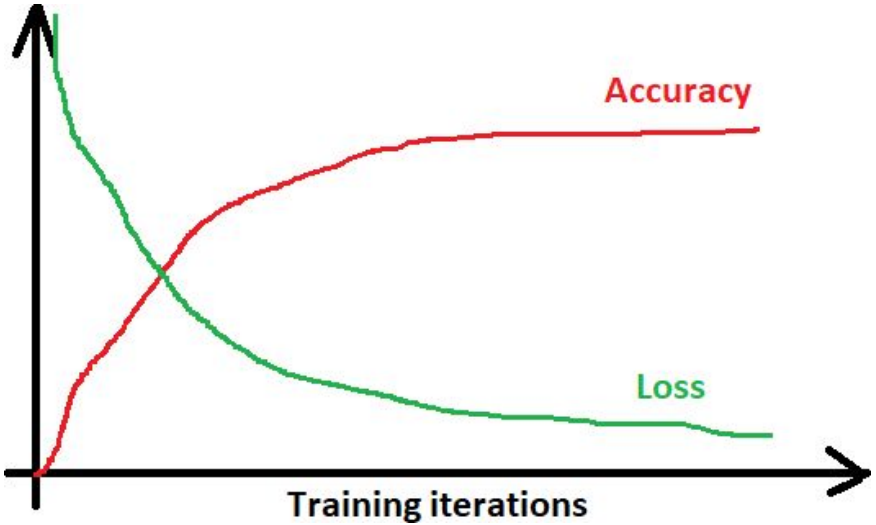
Loss and accuracy

- Training accuracy saturates to a maximum
- Training loss saturates to a minimum
- Loss is a measure of error



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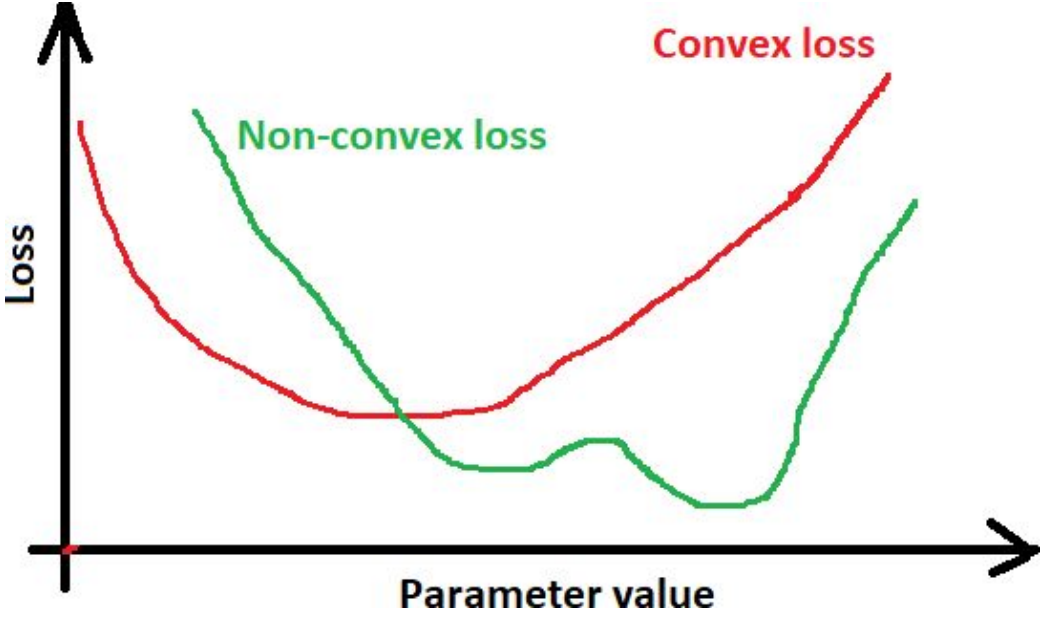
Convex vs. non-convex loss



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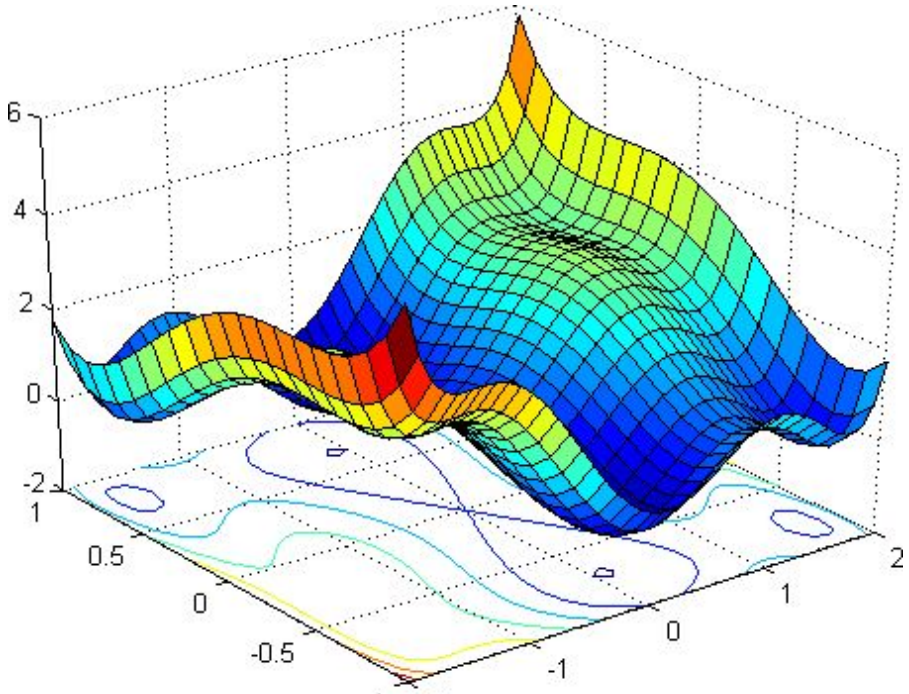
Non-convex loss can have multiple minima



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More about loss function

- Choice of loss function depends on:
  - Desired output type: continuous or categorical?
  - Predicted output type: continuous or categorical?
  - Goal: supervised or unsupervised?

- Loss function over a set is the average of loss over each sample in the set.

- Loss function over the validation set is the most important thing to monitor during training.

- High training loss means under-fitting.

- Large gap between training and validation losses means over-fitting

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Examples of loss functions

- Regression with continuous output
  - Mean square error (MSE), log MSE, mean absolute error
- Classification with probabilistic output
  - Cross entropy (negative log likelihood), hinge loss
- Similarity between vectors or clustering
  - Euclidean distance, cosine

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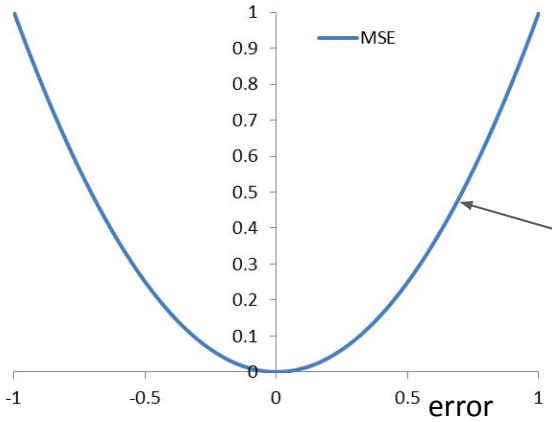
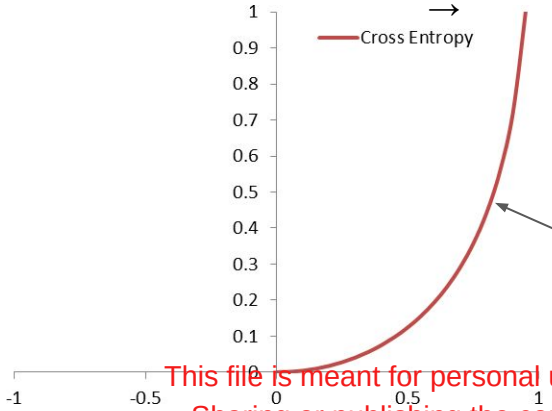
Choice of Loss function



- There are positive and negative errors in classification and MSE is the most common loss function

MSE



- There is a probability of correct class in classification, for which cross entropy is the most common loss function

Cross Entropy

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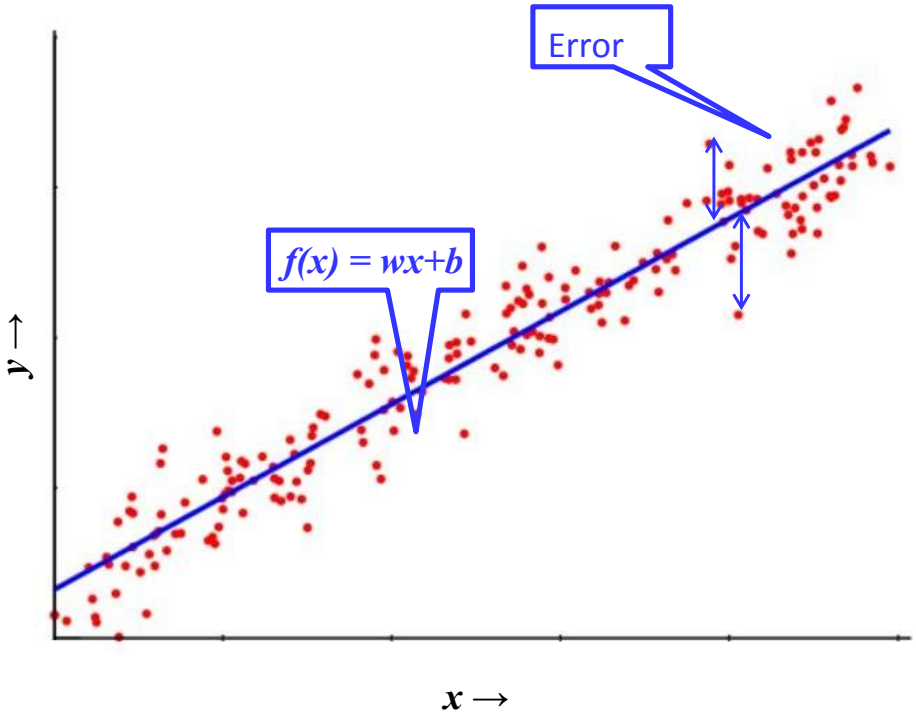
MSE loss for regression



$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

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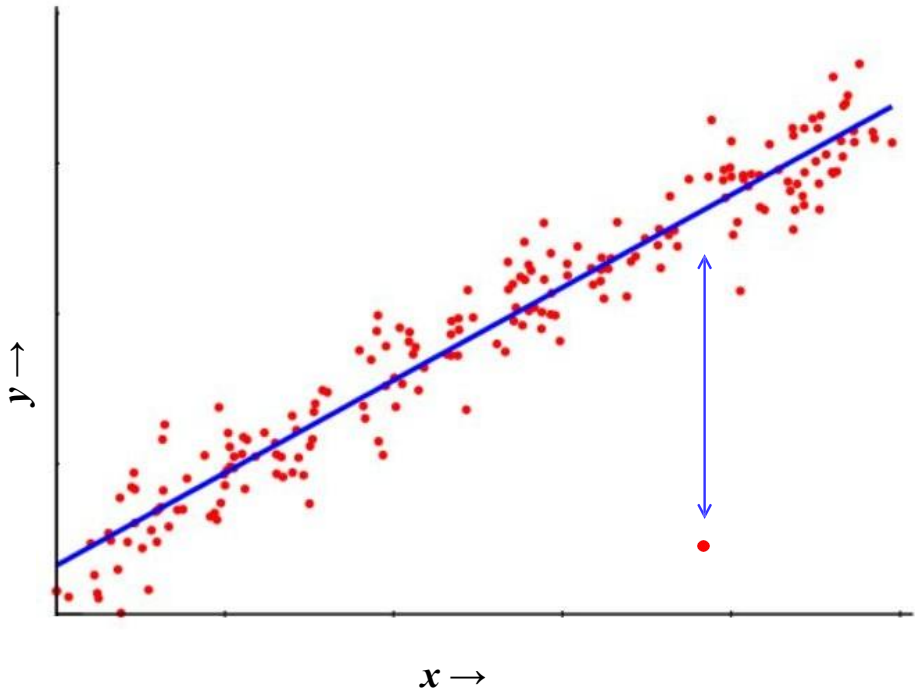
Advantage of MAE



$$\text{MAE} = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n}$$

MAE = mean absolute error

y_i = prediction

x_i = true value

n = total number of data points

MAE loss is less affected by outliers than MSE

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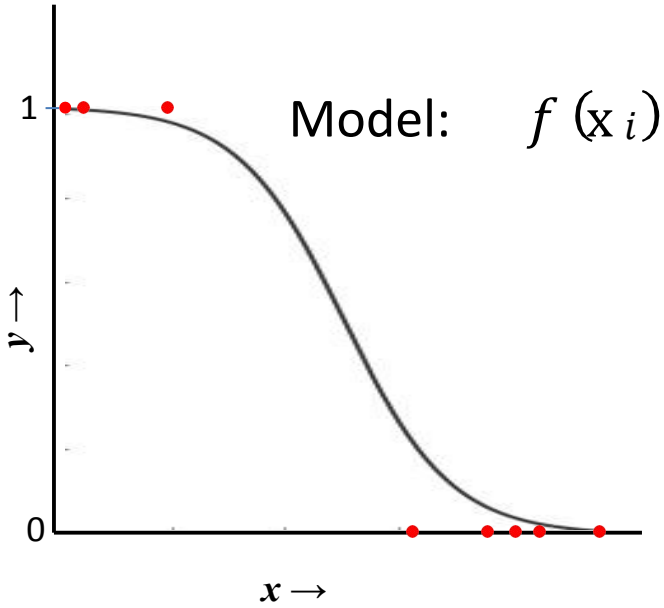
Is MSE appropriate for classification?



Model: $f(x_i) = \dfrac{1}{1+e^{-(wx_i+b)}}$

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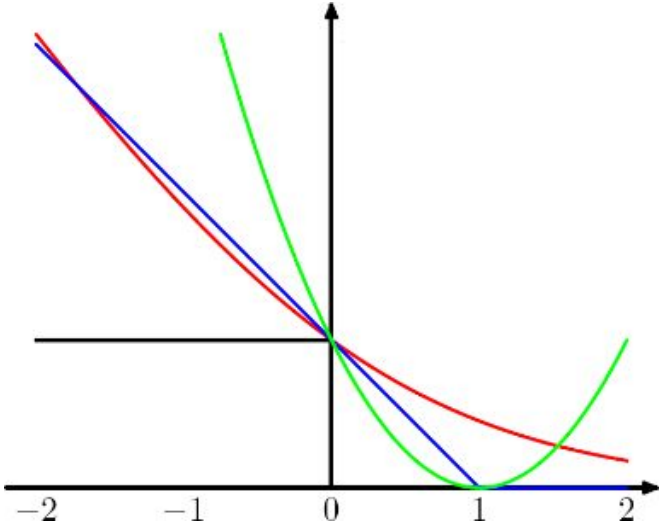
Some loss functions



- Problem: binary classification
- Assumption: desired output is 1
- Notice rate of convergence at different points

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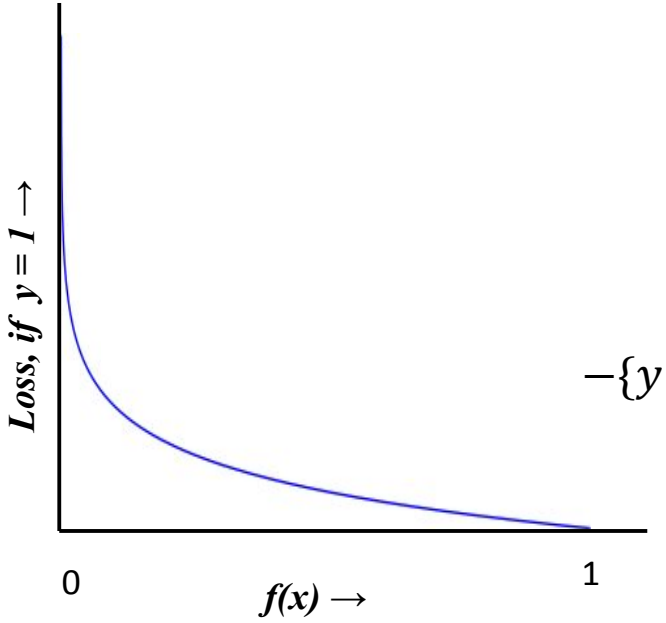
Cross entropy loss: classification



- How much does one (estimated) probability distribution *q(x)* deviates from another (real) *p(x)*
- KL-divergence of *q(x)* from *p(x)*
- For binary classification:

$$-\{y_x * \log f(\ ) + (1-y) * (\log - f(x))\}$$

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Types of Loss Function

(Classification)

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Loss Function for Classification

- Categorical Cross Entropy
- Sparse categorical Cross Entropy
- Binary Cross Entropy

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Loss Function for Classification (Continued)

- **Categorical Cross Entropy**
  - It is a loss function used in multi-class classification and the target labels are always one-hot encoded.
  - It is well suited for classification tasks so that one example can belong to one class with probability 1 and to the other with probability 0.
  - For example, In MNIST digit recognition, the model gives higher probability prediction for the correct digit and lower probabilities for the other digits.

- **Sparse Categorical cross entropy**
  - It is a loss function similar to categorical cross entropy but here the labels are provided as integers rather than being one-hot encoded.
  - An advantage of using sparse categorical cross entropy is that it saves the memory time as it simply uses a single integer rather than a whole vector.

- **Binary Cross Entropy**
  - It is mainly used in problems related to binary classification.
  - It compares each of the probability predicted to the output class which can be either 0 or 1.

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Types of Loss Function

(Regression)

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Loss Function for Regression

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Huber Loss

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Loss Function for Regression (Continued)

- **Mean Squared Error (MSE)**
  - It is calculated as the average of the squared differences between actual and predicted values.
  - The result is always positive and the case with no error will always have MSE=0
  - As it is sensitive to outliers so it is used so MSE will be great for ensuring our trained models has no outlier predictions with huge errors as it implies larger weights on outliers due to the square function

- **Root Mean Squared Error (RMSE)**
  - It is similar to MSE with a root function over it.
  - Here as the errors are squared before the average is taken so RMSE always gives a high weight to large errors.

Both MSE and RMSE have a range from 0 to ∞. The model is said to be good if the value of these errors are low.

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- **Mean Absolute Error (MAE)**
  - It is the sum of the absolute difference between the actual and the predicted values.
  - It is robust to outliers.
  - Its range even lies between 0 and ∞.

- **Huber Loss**
  - It is a loss function used in robust regression as it is less sensitive to outliers in data than the squared error loss.
  - It is differentiable at 0. It is basically absolute error which becomes quadratic when error is small.
  - It combines the good properties of MSE and MAE to yield the best results.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}|$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

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Back Propagation and Learning rate

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Back Propagation

1. Back propagation is the process of learning that the neural network employs to re-calibrate the weights and bias at every layer and every node to minimize the error in the output layer

2. During the first pass of forward propagation, the weights and bias are random number. The random numbers are generated within a small range say 0 – 1

3. Needless to say, the output of the first iteration is almost always incorrect. The difference between actual value / class and predicted value / class is the error

4. All the nodes in all the preceding layers have contributed to the error and hence need to get their share of the error and correct their weights

5. This process of allocating proportion of the error to all the nodes in the previous layer is back propagation

6. The goal of back propagation is to adjust weights and bias in proportion to the error contribution and in iterative process identify the optimal combination of weights

7. At each layer, at each node, gradient descent algorithm is applied to adjust the weights

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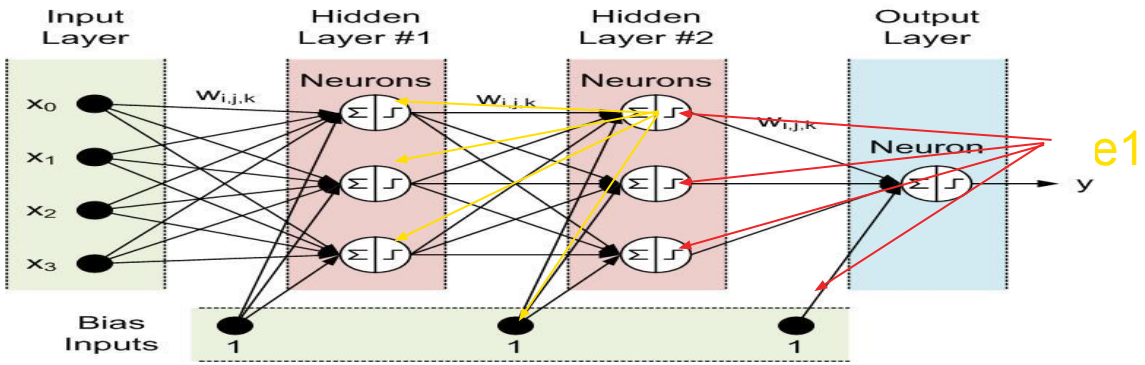
Back Propagation



1. Error in output node shown as e1, is contributed by node 1 ,2 and 3 of layer 2 through weights w(3,1), w(3,2), w(3,3)

2. Proportionate error is assigned back to node 1 of hidden layer 2 is (w(3,1) / w(3,1) + w(3,2) + w(3,3)) * e1

3. The error assigned to node 1 of hidden layer 2 is proportionately sent back to hidden layer 1 neurons

4. All the nodes in all the layers re-adjust the input weights and bias to address the assigned error (for this they use gradient descent)

5. The input layer is not neurons, they are like input parameters to a function and hence have no errors

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Optimization

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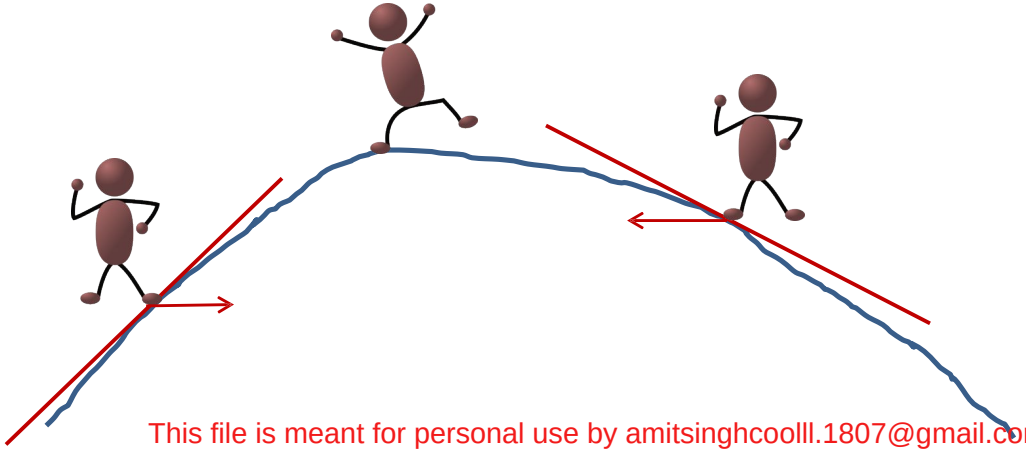
Gradient descent

- If you didn't know the shape of a mountain
- But at every step you knew the slope
- Can you reach the top of the mountain?



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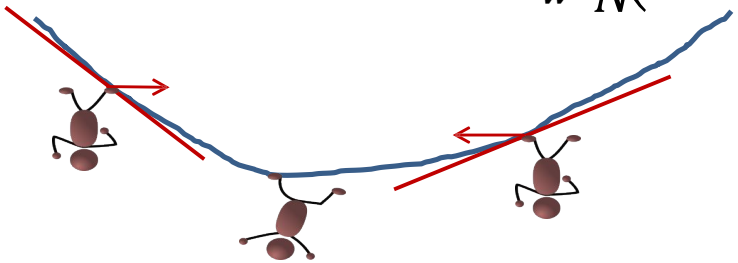
Gradient descent minimizes the loss function

- At every point, compute
  - Loss (scalar): $l_i(\boldsymbol{w})$
  - Gradient of loss with respect to weights (vector):

$$\nabla_{\boldsymbol{w}} l_i(\boldsymbol{w})$$

  - Take a step towards negative gradient:

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \eta \nabla_{\boldsymbol{w}} \left(\frac{1}{N} \sum_{i=1}^{N} l_i(\boldsymbol{w}) \right)$$



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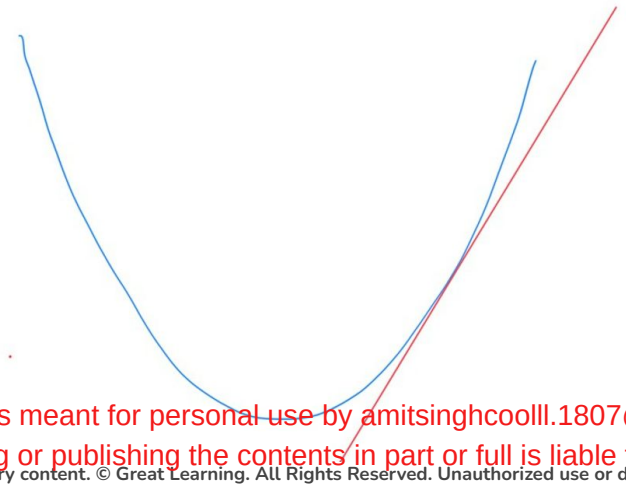
Role of step size and learning rate

- Tale of two loss functions
  - Same value,
  - Same gradient (first derivative), but
  - Different Hessian (second derivative)
  - Different step sizes needed
- Success not guaranteed



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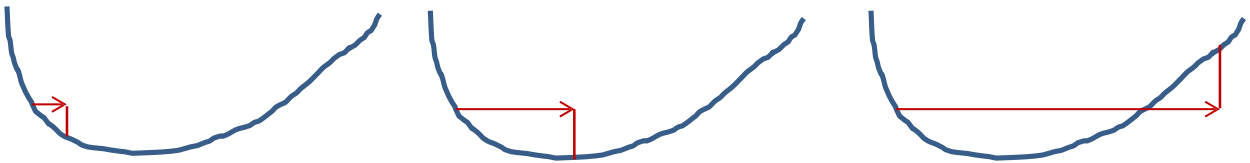
Step size (Continued)

- The perfect step size is impossible to guess. Goldilocks finds the perfect balance only in a fairy tale



- The step size is decided by learning rate η and the gradient

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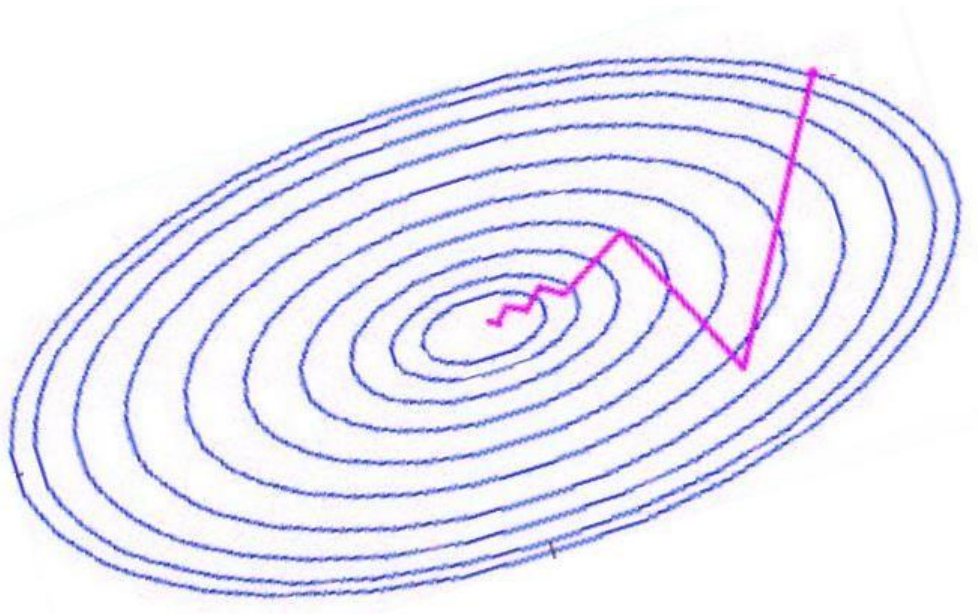
Unfolding step-size in multiple dimensions



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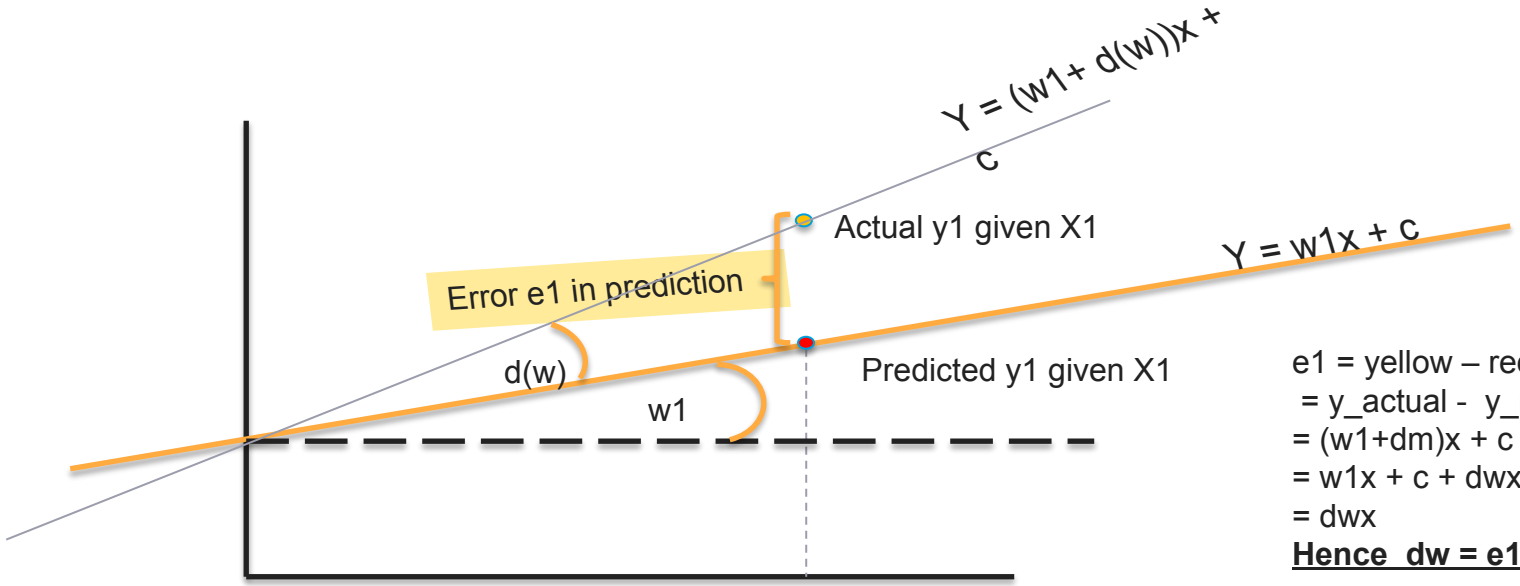
Relation between error and change in weights

Since part of neuron function is linear equation (before applying the non-linear transformation), the error at each neuron can be expressed in terms of the linear equation.



e1 = yellow – red
= y_actual - y_pred
= (w1+dm)x + c - (w1x + c)
= w1x + c + dwx – w1x -c
= dwx
Hence dw = e1 / x

The change required in m (dw) is e1/x. However, change required w.r.t another data point may be different. To prevent jumping around with dw, we moderate the change in W by introducing a **learning rate l** . Hence dw = l(e1/x)

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Number of Samples for each update

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \eta \nabla_{\boldsymbol{w}} \left(\frac{1}{N} \sum_{i=1}^{N} l_i(\boldsymbol{w}) \right)$$

- Vanilla gradient descent: use samples in a fixed sequence:
  $\boldsymbol{w} \leftarrow \boldsymbol{w} - \eta \nabla_{\boldsymbol{w}} l_i(\boldsymbol{w})$
- Stochastic gradient descent: Choose a random sample for each update. In practice, we make random mini-batches based on computational resources
  - Divide N samples into equal sized batches
  - Update weights once per batch
  - One epoch completes when all samples used once
- Batch gradient descent: use all samples, and update once per epoch for all samples

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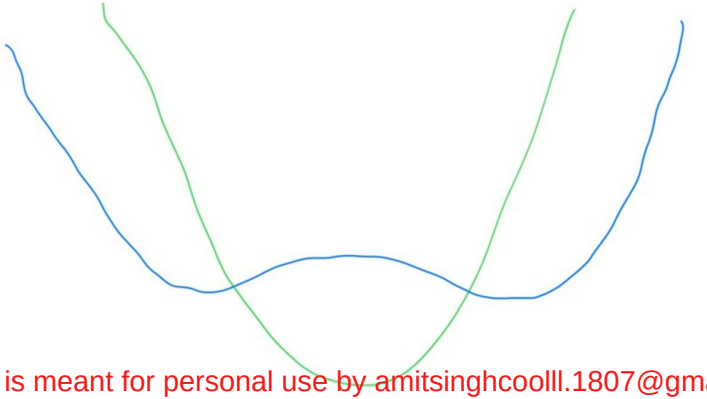
Loss of different sets of samples

- Different mini-batches (or samples) have their own loss surfaces
- The loss surface of the entire training sample (dotted) may be different
- Local minima of one loss surface may not be local minima of another one
- This helps us escape local minima using stochastic or batch gradient descent
- Mini-batch size depends on computational resource utilization



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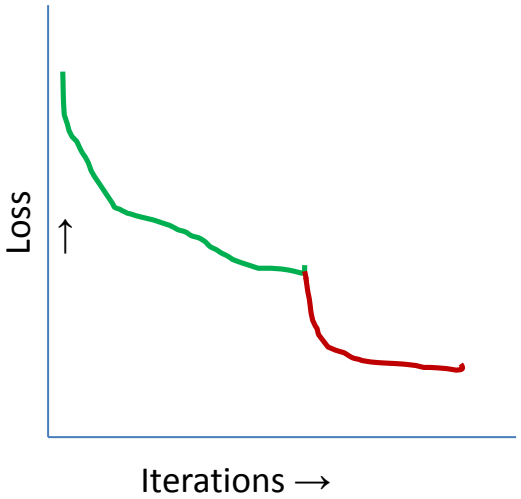
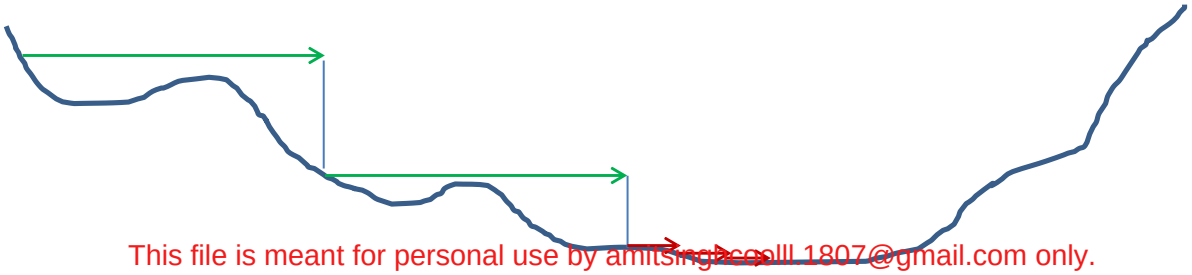
Learning rate decay

- Initially use large learning rate: further from the global minima, we need to rapidly converge
- Later, reduce the learning rate: Closer to the solution we need to start fine-tuning with smaller steps





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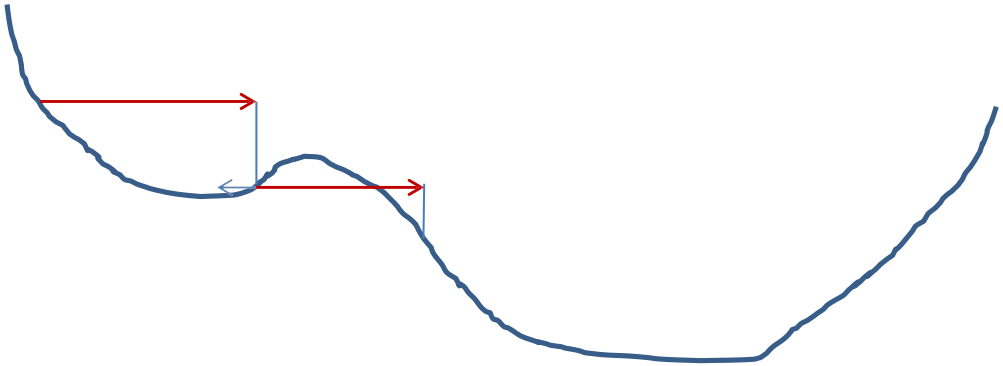
Momentum

- Momentum means using the memory of previous step to build up speed or to slow down with forgetting factor α; $0 \leq \alpha < 1$



$$\Delta \boldsymbol{w}^{(t)} = \alpha\, \Delta \boldsymbol{w}^{(t-1)} - \eta\, \nabla_{\boldsymbol{w}} L(f_{\boldsymbol{w}}(\boldsymbol{X}), \boldsymbol{y}))$$

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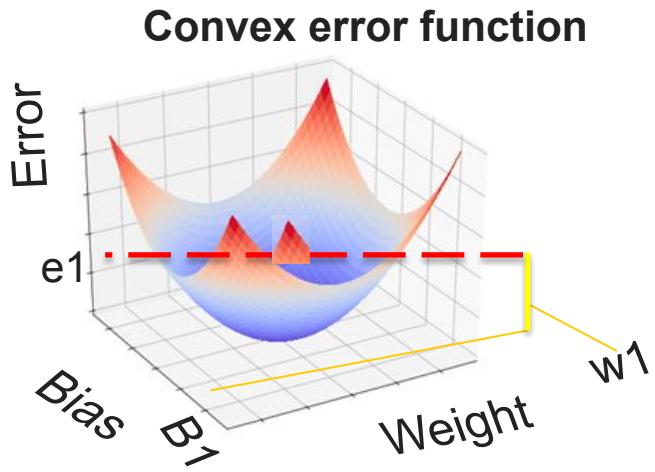
Gradient Descent

The challenge is, all the weights in all the inputs of all the neurons need to be adjusted. It is not manually possible to find the right combination of weights using brute force. Instead, the neural network algorithm uses a learning function called gradient descent.



Note: in 3D error surface can be visualized as shown but not in more than 3 dimensions

1. A random combination of bias B1 and input weights W1 (showing only one as more than one is not possible to visualize)

2. Each combination of W1 and B1 is one particular linear model in a neuron. That model is associated with proportionate error e1 (red dashed line).

3. Objective is to drive e1 towards 0. For which we need to find the optimal weight (Woptimal) and bias (Boptimal)

4. The algorithm uses gradient descent algorithm to change bias and weight form starting values of B1 and W1 towards the Boptimal, Woptimal.

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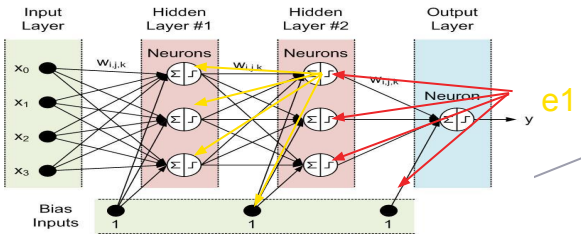
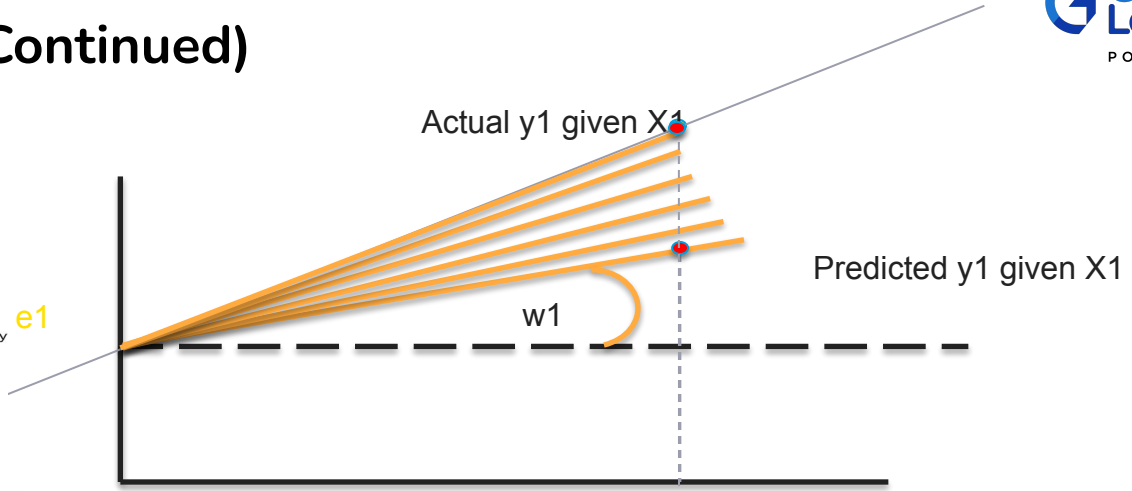
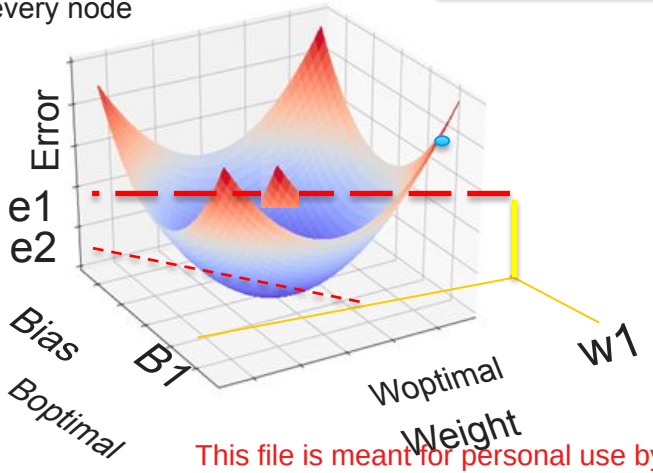
Gradient Descent (Continued)





Happens at every node



Least error E2 is at the global minima of the convex function which only one unique combination of weight (woptimal) and bias (boptimal) will fetch us.

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Gradient Descent (Continued)

- Let target value for a training example X be y i.e. The data frame used for training has value X,y
- Let the model (represented by random m and c) predict the value for the training example X to be yhat
- Error in prediction is E = yhat – y. If we sum all the errors across all data points, some will be positive some negative and thus cancel out
- To prevent the sum of errors becoming 0, we square the error i.e. E = (y – yhat)^2. Note: in squared expression, y – yhat or yhat – y mean the same

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Gradient Descent (Continued)

- Sum of (y – yhat)^2 across all the X values is called SSE (Sum of Squared Errors)

- Using gradient descent (descend towards the global minima). Gradient descent uses partial derivatives i.e how the SSE changes on slightly modifying the model parameters m and c one at a time

d(E) / d(m) = d(sum(yhat – y)^2) / d(m)
d(E) / d(c) = d(sum(yhat – y)^2) / d(c)

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Gradient Descent (Continued)

- Gradient descent is a way to minimize an objective function / cost function such as Sum of Squared Errors (SSE) that is dependent on model parameters of weight / slope and bias.

- The parameters are updated in the direction opposite to the direction of the gradient (direction of maximum increase) of the objective function.

- In other words we change the values of weight and bias following the direction of the slope of the surface of the error function down the hill until we reach minima.

- This movement from starting weight and bias to optimal weight and bias may not happen in one shot. It is likely to happen in multiple iterations. The values change in steps.

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Gradient Descent (Continued)

- The step size can be influenced using a parameter called Learning Rate. It decides the size of the steps i.e. the amount by which the parameters are updated. Too small learning step will slow down the entire process while too large may lead to an infinite loop.

- The mathematical expression of gradient descent.

learning step

$$\theta = \theta - \eta \cdot \nabla_\theta J(\theta)$$

Update Model parameter at e2

Old Model parameter at e1

Gradient descent with learning step

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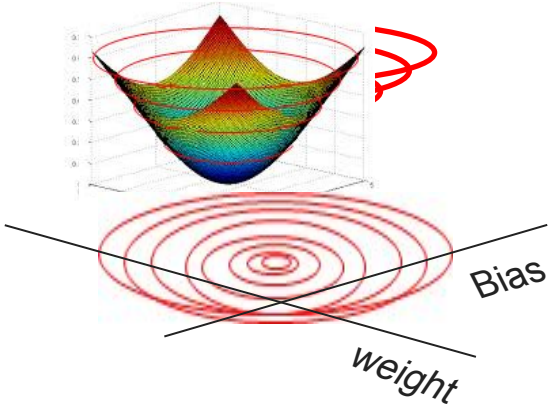
Gradient Descent (Continued)

Transform our error function (which is a quadratic / convex function) into a contour graph. Gradient is always found on the input model parameters only



1. Every ring on the error function represents a combination of coefficients (m1 and m2 in the image) which result in same quantum of error i.e. SSE

2. Let us convert that to a 2d contour plot. In the contour plot, every ring represents one quantum of error.

3. The innermost ring / bull's eye is the combination of the coefficients that gives the lease SSE

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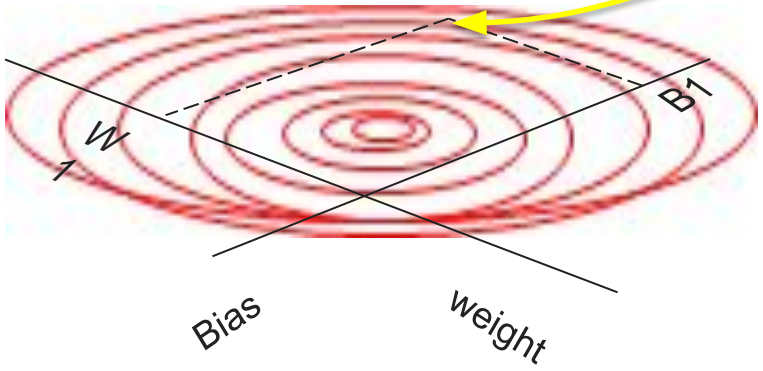
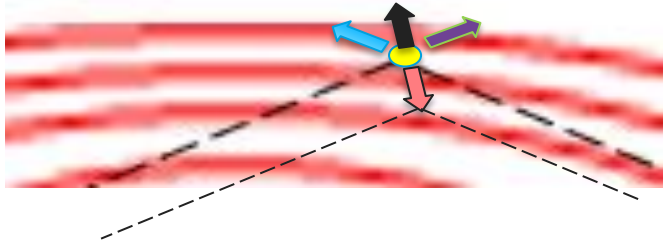
Contour graph

Randomly selected starting point



Gradient vector



About the contour graph –

1. Outermost circle is highest error while innermost is the least error circle
2. A circle represents combination of parameters which result in same error. Moving on a circle will not reduce error.
3. Objective is to start from anywhere but reach the innermost circle

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Gradient Descent Steps –

1. First evaluate dy(error)/d(weight) to find the direction of highest increase in error given a unit change in weight (Blue arrow). Partial derivative w.r.t. to weight

2. Next find dy(error) /d(bias) to find the direction of highest increase in error given a unit change in bias (green arrow). Partial derivative w.r.t. to bias

3. Partial derivatives give the gradient in the given axis and gradient is a vector

4. Add the two vectors to get the direction of gradient (black arrow) i.e. direction of max increase in error

5. We want to decrease error, so find negative of the gradient i.e. opposite to black arrow (Orange arrow). The arrow tip is new value of bias and weight.

6. Recalculate the error at this combination an iterate to step 1 till movement in any direction only increases the error

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Drawbacks of ANN

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Drawbacks of ANN

- ANN models are not much interpretable so it does not give any clue on how it has come up with a solution. Thus it does not make ANN models reliable or trustworthy.

- There is no specific rule for determining the architecture of the network so we mostly use hit and trial method.

- In-order to work efficiently, ANN requires parallel computing power.

- ANN can only work with numerical data so problems or data need to be converted into numerical format before feeding it to ANN.

- Duration of the model to train on the data takes longer.

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